Paths *to* the Rainforests: Ancestral Beliefs and Fertility in Sub-Saharan Africa*

Latest version

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Abstract

This paper examines the demographic consequences of one of the most prevalent belief systems in sub-Saharan Africa. I focus on the impact of a belief system that emphasizes the role of ancestors, who have a strong interest in the continuation of their lineage into which they may be reincarnated. I combine first-hand data, novel ethnographic information, and historical and contemporary surveys to show: 1) a strong, positive relationship between ancestral beliefs and fertility across contexts and time periods; and 2) that this relationship is specifically driven by the motive to continue one's lineage. In a simple model where the total number of children in the lineage is a public good and depends on the kinship system, I show that 1) the effect of ancestral beliefs is driven by patrilineal societies, where the motive to continue one's lineage is stronger; and 2) in groups with ancestral beliefs, a specific pattern of free riding among siblings emerges: in patrilineal societies, fertility decreases with the number of the husband's brothers, whereas in matrilineal societies, female fertility decreases with the number of sisters (but not brothers). These predictions are supported by the data.

Keywords: Fertility, Sub-Saharan Africa, Culture, Supernatural Beliefs, Kinship

JEL Classification: O12, J13, Z12, Z13

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1. Introduction

Fertility is collapsing worldwide, with a total fertility rate that is likely already below replacement. Far from the old popular belief that the world was doomed by high birth rates, population collapse is now understood as a critical challenge with major implications for economic growth and social stability (Bhattacharjee et al., 2024; Maestas et al., 2023). Interestingly, conventional demographic models, which focus almost exclusively on material conditions, underestimate fertility decline in most parts of the world while systematically overestimating it in sub-Saharan Africa (Casterline, 2017).[1]

These limitations in understanding the factors that drive fertility differences across populations suggest room for complementary explanations that consider the importance and evolution of cultural and religious norms, values and practices. Although the relationship between fertility and Abrahamic religions has received considerable attention,[2] we know less about the influence of other prevalent belief systems that transcend religious affiliation. In particular, despite their importance in daily life and their relevance for economic behavior, the study of traditional African religious beliefs within economics has been limited (Le Rossignol et al., 2022; Butinda et al., 2023).

This paper aims to make progress by examining the demographic consequences of one of the most prevalent belief systems in sub-Saharan Africa (henceforth SSA), which emphasizes ancestry and descent and the importance of continuing one's lineage (Caldwell and Caldwell, 1987, 1988, 1990). This contrasts with the widespread and increasing individualism and social atomization observed in most regions of the world in recent decades, and may provide useful insights for understanding demographic change (Santos et al., 2017; Henrich, 2020).

I examine the importance of a belief system that emphasizes the role of ancestors (the spirits of the dead) who influence people's lives and have a strong interest in the continuation of their lineage into which they may be reincarnated (Radcliffe-Brown, 1922; Fortes, 1965; Caldwell and Caldwell, 1987). In these societies, the living are only a frac-

[1] Most of the literature has focused on factors related to economic development: income, infant mortality, education, participation in agriculture, urbanization, and access to health and family planning services (World Bank, 1986; National Reseach Council, 1993). Yet, conditional on these factors, fertility rates remain about one birth higher in sub-Saharan Africa (Zipfel, 2022).

[2] See, for example, Ishak and Gradstein (2022) or Berger and Dasré (2024). For a review, see Götmark and Turner (2023).

tion of a lineage composed of both living members and ancestors, and their common goal is the reproduction of the lineage. To this end, the ancestors bless the fertility of their descendants, bring wealth, health, and children to the family, and reincarnate in new births to the lineage. The resulting belief system works to maintain high fertility by placing great emphasis on the continuation of the family line and reinforcing multigenerational family obligations. In this regard, children are seen as a collective asset that becomes the responsibility of the entire lineage, and a strong moral and spiritual obligation to have children emerges. High fertility is rewarded (both socially and spiritually), associated with righteous living and joy, while low fertility is associated with sin and misfortune, showing the disapproval of the ancestors.

I examine these questions in two steps. First, building on a rich demographic and anthropological literature, I use first-hand data that I collected in Southern Benin and newly digitized ethnographic information that I match to both historical and contemporary surveys to examine the relationship between beliefs in the influence of ancestors and fertility across SSA. I find a strong and positive relationship between ancestral beliefs and fertility rates, of almost one additional child in my preferred specification. This relationship holds in different samples, econometric specifications, and time periods. I start by examining a sample of rural farmers in contemporary Benin, where I conducted a survey that accurately captures beliefs in the influence of ancestors. Second, I focus on post-colonial Democratic Republic of Congo, where I digitized detailed ethnographic information on the practice of ancestor worship[3] at the ethnic group level before European colonization. These data, which is composed mainly of urban migrants, allows me to examine the correlation between beliefs in ancestors and fertility in the spirit of an epidemiological approach. Finally, I show that these results generalize to the wider context of sub-Saharan Africa by using individual-level information on contemporary supernatural beliefs from 25,000 respondents distributed across 19 SSA countries, and that the link between beliefs in ancestors and fertility exists as well in the oral traditions of preindustrial societies, even at the global level.[4]

[3]I will use the term "*ancestor worship*" as a synonym of beliefs in ancestors throughout the paper. However, there is controversy over the suitability of the term ancestor worship in the African context (for further discussion, see Kenyatta (1965) or Grande (2024)).

[4]As a complementary result, I also show a negative correlation between beliefs in ancestors and contraceptive use, once supply-side factors are taken into account (by comparing people living in the same place but holding different beliefs (and therefore facing the same supply-side constraints related to access

The main concerns in the first part of the paper are reverse causality and omitted variable bias.[5] To mitigate these concerns, in the second part of the paper, I develop a simple model of fertility that captures the main mechanism behind the positive relationship between ancestral beliefs and fertility – the strong motive to continue one's lineage – and provides specific predictions that are hard to account for by alternative explanations. In the model, the total number of children in the lineage is a public good because they continue the family line. However, whether one's children belong to one's lineage depends on whether the kinship system is patrilineal or matrilineal, due to the asymmetry in marital allegiances (Fox, 1983).[6] The model makes two main predictions.

A first prediction is that the influence of ancestral beliefs on fertility should be quantitatively stronger in patrilineal than in matrilineal societies, since the motive to continue the family line is stronger. This is because in matrilineal societies a man's children do not continue his lineage, but only that of his wife. In contrast, in patrilineal societies, since wives integrate the lineage of their husbands, children continue the lineage of both parents. Second, there is very specific free-rider behavior among siblings that depends on the kinship system. In patrilineal societies with a beliefs in ancestors, fertility decreases with the number of the husband's brothers, since their children also continue the family line (but the husband's sisters and their children belong to a different lineage, that of their husbands). Following the same logic, in matrilineal societies female fertility decreases with the number of sisters (but not brothers), because the children of the wife's sisters belong to the wife's lineage (but the children of the wife's brothers belong to the lineage of their wives).

to contraceptives, health infrastructure etc.).

[5]Reverse causality is less of a concern since beliefs in ancestors are deeply-rooted cultural beliefs, which are unlikely to emerge at the individual level as a response to a shock. Moreover, the results with the historical data from the DRC alleviate this concern since ancestral beliefs are measured at the ethnic group level and are supposed to represent precolonial beliefs.

[6]In patrilineal societies, lineage is traced through male members, while in matrilineal societies lineage is traced through female members. From the men's perspective, the lineage is continued by one's children and by the children of one's brothers in patrilineal societies, and by the children of one's sisters in matrilineal societies. This is because the husband does not belong to the lineage of his wife in matrilineal societies. From the women's perspective, the lineage is continued by one's children and by the children of the husband's brothers in patrilineal societies, and by one's children and the children of one's sisters in matrilineal societies. The asymmetry comes from the fact that women in patrilineal societies do belong to their husbands' lineage. Therefore, one's children always continue one's lineage in patrilineal societies, while children only continue the mother's line in matrilineal societies. See Appendix L for a simple introduction to kinship systems.

I test these predictions and find that: 1) the positive influence of ancestral beliefs on fertility is stronger in patrilineal societies;[7] And 2) there is a negative relationship between one's fertility and the number of family members capable of continue one's lineage with their own children. I show that, relative to patrilineal societies, female fertility decreases with the number of sisters in matrilineal societies with strong ancestral beliefs. Moreover, this relationship disappears when we look instead at the number of brothers, or when we examine societies without a belief in ancestors. On the contrary, I show that compared to matrilineal societies, fertility decreases with the number of men aged 15-49 living in the household, but only in patrilineal societies with strong ancestral beliefs.[8]

Overall, these patterns are consistent with an environment in which the motive to continue the family line drives fertility upward and are difficult to explain by alternative hypotheses. Moreover, they highlight the strong link between belief systems and prevailing social structures. In this direction, I examine some additional implications of my findings to further validate the theory at hand. In particular, I examine how the relationship between ancestral beliefs and fertility depends on two different types of social organization. Because ancestor worship is a lineage-level cult, I examine whether ancestral beliefs disappear as the lineage-based structure of societies becomes less salient. I make use of the distinction between lineage or kin-based societies and age-based societies (Moscona and Seck, 2024), and show that the relationship between ancestors and fertility is stronger in lineage-based societies than in age-based societies, where social units are mainly organized "horizontally" rather than "vertically". Overall, my results show that fertility decisions are made within the larger framework of the family, clan, or lineage, and are therefore not driven solely by the interests of an individual (or couple), but rather by a much broader collective interest. In this regard, belief systems are power-

[7]Similarly, I show that the positive relationship between patrilineality and fertility only exists in societies with ancestral beliefs, which reduces concerns about the results being driven by differences in the characteristics of matrilineal and patrilineal societies that are not related to the motive to continue the family line (unless these unobserved differences be correlated with the existence of ancestral beliefs).These results also help us to better understand the positive relationship between patrilineality and fertility suggested in other contexts (BenYishay et al., 2017; Okafor et al., 2021; Fontenay et al., 2024). I provide supportive evidence that group membership (parents and children belonging to the same lineage or not) is a key mechanism for understanding the relationship between kinship structure and fertility.

[8]Unfortunately, the DHS does not include information on siblings for men, so I use the number of men aged 15-59 living in the household as a proxy for the number of men who can continue the respondent's lineage by having children.

ful tools for reinforcing (or diluting) the cooperative ties that underpin the core of child rearing and may have contributed to maintaining human fertility above replacement levels since the emergence of our species (Boyd and Silk, 2003; Henrich, 2016; Gurven and Davison, 2019).

This paper contributes to several strands of the literature. First, I contribute to existing analyses on the role of culture and religious beliefs on fertility in sub-Saharan Africa (e.g., Ishak and Gradstein (2022), Berger and Dasré (2024), or Götmark and Turner (2023) and Church et al. (2023) for reviews).[9] [10] In particular, I build on an extensive qualitative literature in demography and anthropology that linked the importance of descent in many African societies to their belief systems and certain aspects of their social organization, which focused on maintaining the family line and reverence for ancestors (Caldwell and Caldwell, 1985, 1987, 1988, 1990; Lesthaeghe, 1989; Caldwell et al., 1992). To the extent of my knowledge, I am the first to quantify the influence on fertility of one of the most prevalent belief systems in the context of sub-Saharan Africa.

Second, my findings contribute to the literature on the relationship between culture and social structures, and its influence on fertility (e.g., Caldwell and Caldwell 1987). More than 70 years ago, Lorimer (1954) investigated the relationship between fertility and unilineal descent groups. However, this literature has paid little attention to the religious underpinnings of social structures. In this paper, I demonstrate the importance of religious and cultural beliefs that emphasize the significance of perpetuating family lineage in societies where patrilineal descent is prevalent, thereby highlighting the interconnected evolution of religious and social systems. Additionally, my findings contribute to the literature on how kin influence one's fertility and how family members interact strategically to raise children (see, for example, Sear 2017; Rossi 2018). I show that fertility decisions are influenced by the larger framework of the family, clan, or lin-

[9] Most often, these papers compare the fertility trajectories of major religious groups, mainly Christians and Muslims, relying on the Demographic and Health Surveys. The general consensus is that Muslims have higher fertility than Christians, while followers of African Traditional Religions and Islam have similar fertility levels. This differential fertility is largely explained by the position of women within households. However, these papers have important limitations. First, conventional demographic surveys such as the DHS are very problematic to study these questions since they under-report the importance and the coverage of African traditional religions and beliefs (which, although more common among followers of African Traditional Religions, are transversal to religious affiliations and to socioeconomic conditions).

[10] This paper is also related to the broad literature on the relationship between fertility and other cultural or social aspects in SSA, such as clan linkages (Bauer et al., 2006), inheritance rules (Fontenay et al., 2024; Sage, 2023), or postpartum sexual abstinence (Bertocchi et al., 2025).

eage (Bauer et al., 2006). Therefore, they are not solely driven by an individual's (or couple's) interests but rather by a much broader collective interest. Specifically, my findings demonstrate that in contexts where the desire to continue the family line increases fertility, there is free-riding behavior regarding fertility among siblings whose children continue the same family line (see, in the context of China, Yang and Spencer 2022).[11]

Finally, I contribute to the broad literature on the importance of cultural characteristics for socioeconomic outcomes (e.g., Luttmer and Singhal 2011; Alesina et al. 2013; Alesina and Giuliano 2015). Regarding fertility, Fernández and Fogli (2009) study how fertility rates in the country of origin of second-generation American women have a positive and significant effect on their number of children. In particular, this paper adds to a growing literature on the effects and consequences of traditional supernatural beliefs in Africa (Gershman, 2016, 2022; Stoop et al., 2019; Alidou and Verpoorten, 2019; Stoop and Verpoorten, 2020; Sievert, 2023; Nunn and Sanchez de la Sierra, 2017; Igboin, 2022; Butinda et al., 2023). However, most of the papers related to aspects of African Traditional Religions have focused on the consequences and nature of "witchcraft", while other prevalent belief systems have received little attention, such as beliefs in ancestors.[12] Most of the effort dedicated to the recent study of ancestor worship comes from cultural anthropology. Indeed, no paper has examined the economic and social consequences and implications of beliefs in ancestors in the context of sub-Saharan Africa from a quantitative perspective, despite being recognized that these beliefs are prevalent and relevant to economic behavior.[13]

The remainder of the paper is organized as follows. Section 2 introduces the context

[11]Similarly, my findings underscore the importance of considering men's beliefs and motivations when examining reproductive patterns. It is surprising, then, that although it is widely recognized that men in sub-Saharan Africa have greater bargaining power than their spouses and often make fertility decisions alone, men's fertility has largely been overlooked in demographic research (Coleman, 2000; Dudel and and, 2019; Schoumaker et al., 2024). I demonstrate that men's individual incentives to have childrenfor example, differences in the strength of the motive to continue their lineage depending on the kinship systemare key to understanding fertility.

[12]One exception is the ongoing work of Lowes et al. (2025), where they examine how traditional beliefs that emphasize connection with ancestors interact with local politics.

[13]There are studies investigating the role of ancestor worship practices and fertility in other contexts, such as China. For instance, Zhang (2024) exploits a natural experiment, the Kuomingtangs Retreat to Taiwan, which resettled approximately one million Chinese in Taiwan between 1945 and 1954, to show that ancestor worship (a cultural feature of the Chinese population) contributed to the transmission of son preference and high fertility rates. In the case of contemporary China, Hu and Tian (2018) find a positive correlation between current ancestor worship practices and childbearing and marriage outcomes.

and describes why there exists a close relationship between beliefs in ancestors and fertility. Section 3 presents the different datasets and variables used in the empirical analysis. Section 4 describes the different specifications and depicts the main results. Section 5 proposes a simple theoretical framework to better understand the empirical regularities. Finally, Section 6 tests some of the theoretical predictions and shows that the motive to continue one's lineage is key to explain the empirical results. Section 7 concludes.

2. Traditional religion and ancestorism in Africa

The belief in the ancestors is usually identified as one of the main components of African Traditional Religions (Idowu, 1973).[14] According to Asante (2009), "ancestors are those who once lived in human society and, having fulfilled certain conditions, are now in the realm of the spirits".[15] The belief in ancestors is considered as a crucial element of the traditional belief system of many African societies (Fortes, 1965). In contrast to cults of the dead, anthropologists have observed that the key characteristic of ancestors is the belief that the dead person continues to influence society (Radcliffe-Brown, 1922; Kopytoff, 1971).

Ancestors are seen as dispensers of both favor and misfortune, and their powers are most often used to ensure the survival of the lineage by contributing to the unification of families and people, by bringing health and children to the family, or by offering protection from disease, evil, or enemies. However, ancestors are also believed to be the source of illness, misfortune, or disruption in the lives of their descendants, with their ultimate power being the curse that brings sterility and child death (Fortes, 1965;

[14]An exhaustive description of African Traditional Religions (ATR) is beyond the scope of this paper (see Idowu (1973), Mbiti (1975), Opoku (1993) or, more recently, Olupona (2014) for detailed descriptions). According to Idowu (1973), five elements go into the making of ATR: belief in God, belief in the divinities, belief in spirits, belief in the ancestors, and the practice of magic and traditional medicine. Of course, the weight of each of these components varies from society to society, ranging from very prominent in some areas to virtually absent in others.

[15]The question of who established these criteria is not entirely clear (Igboin, 2022). However, some of these conditions, often of a moral and social nature, are common to different societies. For example, one must have lived an exemplary life by the standards of the community, respected the elders, married, and had children. In addition to having children, an additional criterion often mentioned is to have at least one son who will worship him, while having only female children may not be enough to become an ancestor. On the other hand, death should be natural and at a mature age, and proper burial rites must have taken place.

Caldwell and Caldwell, 1987; Ezenweke, 2008).

A final aspect relates to the prevalence of the belief in ancestors in sub-Saharan Africa as compared to other parts of the world. In Appendix A, I discuss reasons why this belief system is more widespread nd influent in the context of sub-Saharan Africa than in any other region.[16]

2.1. *Implications of Ancestral Beliefs on Fertility in SSA*

Why are beliefs in the ancestors closely related to fertility behavior in SSA? To answer this question, it is important to emphasize that ancestor worship belongs to the realm of kinship and lineage structures (Fortes, 1965; Gong et al., 2021). As Fortes (1965) notes, ancestor worship is a lineage cult in many African societies – that is, a cult of the basic politico-jural unit of many African societies, rather than of the domestic sphere.[17] Similarly, Turaki (2000) mentions that "the ancestors are the most powerful, basic and primary component of the *kinship system* of an African community". In fact, ancestor worship is the extension of the core elements of the social structure of many African societies – ancestry, kinship and descent relations – to the supernatural sphere (Fortes, 1965; Caldwell and Caldwell, 1985).[18]

The belief in the intervention of ancestral spirits in everyday life and the need for descendants to ensure the survival of the lineage is continuous with the social structure. The lineage is thus understood as "a group of descent stretching back infinitely and with an enormous spiritual investment in reaching indefinitely into the future" (Caldwell and Caldwell, 1987). Only a portion of the entire lineage is alive at any one time, and its extension into the future is the central concern of both living lineage members and ancestors. Ancestors maintain their connection to the lineage after death, and each

[16]Some authors (e.g.,Kenyatta (1965)) prefer the term "*communion with the ancestors*" in the African context rather than ancestor worship to distinguish between the belief system of many African societies and the cult of the ancestors that exists in other regions, particularly Southeast Asia. Similarly, Grande (2024) notes that "Ancestorism" is not always a matter of ancestor worship. It is most often a matter of ancestral spirits helping sustain the living family while being sustained by it in return (through rituals). For simplicity, I will not make distinctions between these terms.

[17]This is easily seen in matrilineal societies, where it is the mother's brother who becomes an ancestor, rather than the father himself.

[18]The fact that Traditional African Religions symbolize the key elements of social organization led some authors to conclude that "to a great extent, African religion is essentially the reproduction of the lineage" (Caldwell and Caldwell, 1987).

new birth into the lineage is a way for an ancestor to return through reincarnation (Mbiti, 1975).[19] Qualitative evidence from first-hand data collected in Benin confirms the close link between ancestors and the importance of lineage extension: 95% of those who believe in ancestors ($n = 1788$) also believe that their ancestors are deeply concerned about the continuation of their lineage.

The emphasis on ancestry and descent and the consolidation of lineage - that is, the succession of generations - as a key element of these societies have important demographic consequences. For example, referring to societies in East Africa, Molnos (1973, p.129) mentions that continuing the lineage and commemorating ancestral spirits is one of the precise reasons for wanting "as many children as possible" since "*lineages competed by producing numerous offspring for the favor of ancestors on whom their welfare was deemed to depend*".

The argument presented above can be interpreted as an extension of the well-known "old-age security motive", whereby people's needs for old-age support raise the demand for children.[20] In vertical systems of transmission where the continuation of the family line is of central importance due to the influence of ancestors, the question of "security" has two different facets (Goody, 1973). In addition to the standard security in old age, there is the security in the after-life which drives the demand for children in the same way as the former, and which can be obtained by making provision for the continuity of the family estate.

Maintaining the lineage remains one's responsibility, whether in the human world or in the afterlife. To this end, ancestors bless their progeny's fertility, provide wealth,

[19]For example, among the Fon of Benin, the Yoruba of Nigeria or the Beng people of Côte d'Ivoire, new children born into the family are believed to be reincarnated when an old person has recently died, and children are often named to identify the ancestors reborn in their form (Caldwell and Caldwell, 1987; Osanyìnbí and Falana, 2016). Ancestors may return in more than one child in a family. For example, (Idowu, 1973) notes for the Yoruba that "it is believed that [ancestors] reincarnate not only in one grandchild or great-grandchild, but also in several contemporary grandchildren and great-grandchildren who are brothers and sisters and cousins, aunts and nephews, uncles and nieces, *ad infinitum*". Similarly, in the context of urban South Africa, Anderson (1993), p.27, reports: "One respondent said that in 1986 she had a dream in which she saw she was pregnant. Someone took her to a big stone (probably a gravestone) on which was written the name "Isaac." The following day she inquired from an older family member, who said that Isaac was a grandfather who had died many years previously. A month later the respondent fell pregnant and a baby boy was born, whom she had to call Isaac. She then prayed and thanked the ancestors for their gift of the child. The child thereby, following traditional custom, received the "ancestor spirit" of the deceased ancestor Isaac."

[20]See Lambert and Rossi (2016), Rossi and Godard (2022), or Sage (2023) for evidence of the old-age security motive for fertility in the African context.

health and children to the family, and reincarnate into new births to the lineage (Grande, 2024; Olupona, 2014).[21] It is therefore not surprising that the most common use of ancestral powers is related to lineage survival and the well-being of the kinship group. Moreover, high fertility, since shows the approval of ancestors, is morally associated with joy, recognition, and right living. On the other hand, low fertility or infertility are associated with misfortune, sin, mistreatment, or marginalization, showing the disapproval of ancestors. These associations are all the more important for human behavior because in societies organized hierarchically by age, ancestors are usually the most respected and venerated figures of the community, holding even greater authority than elders.

3. Data

To study the role of traditional religious and cultural beliefs in the influence of ancestors on fertility, I rely on several datasets from different contexts and time periods, including original survey data and novel digitized ethnographic information. These datasets, which span more than a century, allow me to test the relationship between ancestral beliefs and fertility in different contexts, at different levels of aggregation, and with different levels of precision and representativeness.

3.1. First-hand data from contemporary Benin

I collected data from a sample of 943 households living in rural areas of Southern Benin. The respondents were randomly selected, subject to inclusion criteria (e.g., having more than 0.5 hectares of land), as part of a large project to support women in pineapple production.[22] The main survey was conducted in 2024 in a standard face-to-face setting. Respondents were randomly assigned to enumerators to minimize enumerator bias. Along with a comprehensive set of socioeconomic characteristics and fertility outcomes, I directly asked about traditional beliefs in ancestors. The main variable I use to

[21] These insights may help us to better understand the absence of a quantity-quality (QQ) trade-off in some Sub-Saharan African societies (Alidou and Verpoorten, 2019). One could think, for example, that in addition to have more children, parents would invest more in those children who are believed to be reincarnated ancestors.

[22] Both husband and wife were supposed to be interviewed in each household. However, because the project focused on women, some households only included the woman (71/943) because she is not married or because the husband was not present at the time of the survey (29/872).

measure ancestral beliefs use answers to the question "*do you believe that the spirits of your ancestors have an influence on the events of your life?*"[23][24]

3.2. Zaïre and contemporary Democratic Republic of Congo

West Zaire Surveys (1975-77) and Jan Vansina's Congo's Ethnography.– I use original information from two demographic surveys conducted in the western part of the DRC between 1975 and 1977 (see Appendix F for details of these surveys).[25] The first contains individual-level information on 250,000 individuals in 43,000 households living in seven major cities in the DRC (in tables I refer to this survey as the urban sample). The second includes individual-level information on almost 50,000 individuals in 11,000 households distributed across four regions, and covers both rural and urban areas, excluding the large cities included in the urban survey (in tables I refer to this survey as EDOZA). These surveys contain information on respondents' ethnicity and birth calendars for each woman over the age of 13. Using the self-reported information on ethnicity, I match respondents in the surveys to the ethnicity-level information on beliefs in ancestors present in Vansina (1966)'s book.[26]

Vansina (1966) is the main ethnographic source that I use to measure the prevalence of beliefs in ancestors at the ethnicity-level in the DRC. From this book I have digitized original ethnic-level information on the practice of ancestor worship. I constructed a dummy variable that equals one if it is explicitly mentioned that an ethnic group practices ancestor worship. This ethnographic source has two clear advantages. First, it contains explicit information on the main variable of interest so no proxy is needed. Sec-

[23]An example was provided after each question to clarify its meaning. Regarding the first question: "For example, do you think that if you are lucky in your income-generating activities, it is because your ancestors are behind you? Regarding the second question: "For example, do your ancestors influence the health of your children or protect them from bad events?".

[24]This variable is rather a measure of the *strength* of individuals' traditional beliefs in ancestors. In fact, qualitative interviews revealed that it is possible to answer negatively to this questions while still believe in the *existence* (rather than influence) of ancestors. The estimates are therefore likely to represent a lower bound of the true effect.

[25]Although these surveys have been used by demographers to create statistics at an aggregate level (see, for example, Tabutin (1982) or Shapiro (1996)), the use of the microdata is rather unique (they have only been used in Alvarez-Aragon et al. (2025) and Guirkinger and Villar (2022)).

[26]Cogneau and Dupraz (2015) wonder to what extent the classifications of ethnic groups made by anthropologists have influenced the classifications used by the surveys. In the case of the demographic surveys used here, their codebook explicitly mentions that they adapted the questionnaires to record ethnicity as previously classified by Vansina (1966).

ond, it provides a much more detailed description of the ethnographic landscape of the DRC than traditional datasets such as the Ethnographic Atlas. While the EA records 60 ethnic groups in the territory of the present-day DRC, and about 400 for the entire SSA, this number increases to about 250 in Vansina (1966). Of these, 66% practice ancestor worship. The matching procedure used here has also the advantage of taking migration into account, compared to matching techniques based on GPS coordinates, and will allow me to exploit variations in ethnicity-level beliefs in ancestors among migrants of contiguous ethnicities and same destination. Overall, I am able to match about 60,000 women over the age of 13 with the information provided by Vansina (1966).[27]

DRC Demographic and Health Surveys (DHS).– Finally, I complement the analysis with the two available rounds of the Demographic and Health Surveys available for the Democratic Republic of the Congo (2007-2013), a nationally representative survey that provides detailed information on education, literacy, occupation, religion, fertility preferences, and contraceptive methods. I use data on about 40,000 individuals living in 785 clusters from both the men's and women's questionnaires. To measure ancestral beliefs, I follow the methodology of previous studies (i.e., Lowes, 2022) and use location to match respondents in the DRC DHS with the ethnographic information.[28] To do this, I combine the location of each DHS cluster with a digitized version of the map of ethnicities from Vansina (1966). The matching procedure is shown in Appendix C.

3.3. Sub-Saharan Africa: Pew Research Center's Forum on Religion and Public Life

In an attempt to examine the external validity of the previous analysis performed in two specific contexts, I use individual-level information on contemporary supernatural beliefs from 25.000 respondents in 19 SSA countries, coming from a survey conducted by (Pew Research Center, 2009). I measure the strength of ancestral beliefs by using answers to the following question: "*Do you ever participate in traditional African ceremonies*

[27]This figure represents about 70% of the total number of women over the age of 13 included in the demographic surveys. Of the non-compliant subsample, 30% are individuals born in a country other than the DRC.

[28]Using self-reported ethnicity to match the DHS data with Vansina's information on ancestor worship is less satisfactory because the information on ethnicity in the DHS for the DRC is very limited, and the match of the DRC ethnicities with the Ethnographic Atlas is quite difficult. In fact, only 7 different ethnicities (or groups of ethnicities) are recorded in the DHS (compared to about 130 after matching with Vansina).

or perform special acts to honor or celebrate your ancestors?".[29][30] Appendix D.1 shows the spatial distribution of the main explanatory variables.

3.4. *Berezkin's Folklore and Mythology Catalogue and Ethnographic Atlas*

Finally, to show that the relationship is not driven by factors only affecting the post-independence fertility trajectories of societies, I leverage information from the folklore catalogue constructed by Michalopoulos and Xue (2021), combined with the Ethnographic Atlas (Murdock, 1967). This dataset includes information on the proportion of folkloric motifs in a given oral tradition related to different concepts. These motifs are considered to be the building blocks of oral traditions, representing their characteristics, important experiences, events and images, and are particularly valuable because culture is transmitted from generation to generation through oral traditions such as myths, folklore, stories, or proverbs (Galor et al., 2023).[31] I measure ancestral beliefs in pre-industrial societies by looking at the proportion of folkloric motifs in an ethnic group's oral tradition related to the word *ancestor*, and the importance of fertility by looking at the proportion of motifs related to the word *birth* in an ethnic group's oral tradition.[32]

4. The influence of ancestral beliefs on fertility

4.1. *Evidence from Contemporary Benin*

I start by examining the relationship between beliefs in ancestors and fertility using first-hand data collected among a sample of (mostly married) rural households ($n = 943$) in

[29] I show that my results do not change when using different measures of ancestral beliefs. For example, I construct a dummy variable that equals one if the respondent answers "*yes*" to the question "*Do you believe that sacrificing to spirits or ancestors can protect you from bad things happening?*". Another alternative is to construct a dummy variable equal to one if the respondent answers "*yes*" to the following question: "*Which, if any, of the following do you believe in? Reincarnation*".

[30] As with the first-hand data from Benin, this survey measures ancestral beliefs at the individual level, and therefore does not consider ethnicity as the only vector of cultural transmission, since the analysis here does not rely on ancestral ethnic group's characteristics.

[31] Even in recent times, the presence of ancestors in the culture of many African societies is widespread (not only in their oral and written traditions, but also in their art). For example, Appendix B shows a poem written by the Senegalese writer Birago Diop (1906-1989) in 1960. Appendix B.3 shows the statue of an ancestor from Katanga province. Interestingly, it is a male statue who is pregnant male, highlighting the central importance of the succession of generations.

[32] I also use alternative measures. For example, I show that my results remain unchanged if I use the word *ancestral*, the combination of the words *ancestor* and *worship*, or the word *fertility* instead of *birth*.

southern Benin. This is a valuable context that allows me to disentangle alternative mechanisms: almost everyone's main activity is agriculture (98% of men and 90% of women have agricultural fields), fertility is still very high (the average number of children is about 8 for men and above 5 for women), and the clan-based structure of the society is pervasive (45% of men and 41% of women have received pressure from their extended family or clan to increase their number of children). Beliefs in ancestors are widespread: 43% (31%) of men (women) believe that ancestors have an influence on their lives. Consistent with the qualitative evidence that emphasizes the central role of lineage continuity in this belief system, 96% of those who believe in the influence of ancestors also think that their ancestors are deeply concerned about the survival of their extended family and the continuation of their lineage. The baseline specification, estimated through ordinary least squares, takes the following form:

$$Y_i = \alpha + \beta_1 AncestralBeliefs_i + X_i' \Phi + \epsilon_i \tag{1}$$

Where Y_i is the number of children ever born and $AncestralBeliefs_i$ is a dummmy variable equal to one if the respondent (or the respondent's spouse or both) answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?* The vector X_i' includes an extensive set of control variables.[33] Since men are the main decision-makers regarding fertility in this context, my baseline specification focuses on women but considers the influence of both husband and wife's beliefs.[34]

Table 1 presents the estimates for women while sequentially including different controls. Columns (1)-(4) do not include religion fixed effects, while column (5)-(8) do. Columns (1) and (5) do not include control variables, columns (2) and (6) include only baseline controls (age, education, and access to electricity), and columns (3) and (7) include the full list of controls. Finally, columns (4) and (8) include 1km-buffer fixed effects,

[33]As baseline controls, I always include age, age squared, education, and whether the household has access to electricity. Second, I add additional control variables that help me to disentangle alternative mechanisms: total number of agricultural fields, and whether the household has a TV (as a proxy for wealth). Finally, I include religion fixed effects (five main categories: Catholic, Celestial Church of Christ, Evangelical, Voodoo, and Other, and 1km-buffer fixed effects.

[34]This is the reason why the number of observations drops, since I focus on married women. Appendix G.1 reports the results for women only, for men only, and for men while considering the beliefs of both spouses. Interestingly, the size of the coefficients are slightly higher when both spouses hold strong beliefs in the influence of ancestors, rather than the husband or wife alone. These effects suggest some degree of complementarity between spouses and room for non-negligible importance of women's decision making.

so I compare respondents living very close to each other but holding different beliefs. This specification allows to take into account supply-side factors such as access to contraceptives.[35] Robust standard errors are reported in parenthesis. Finally, I show in the last two rows of the table that these results are extremely unlikely to be fully explained by unobservables, following Altonji et al. (2005) and Oster (2019)'s methodology (see Appendix G.9 for a brief explanation). Both the Altonji ratio and Oster's delta indicate that unobserved confounders would need to be orders of magnitude more influential than the rich set of observable controls I include (i.e., age, education, religion, wealth...) to explain away the estimated effects, reducing concerns about omitted variable bias.

Across all specifications, I find a positive, stable, and statistically significant relationship between beliefs in ancestors and fertility: holding strong traditional beliefs in the influence of ancestors is associated with about 0.7 additional children (13% of the outcome mean). These effects are quantitatively large, bigger in magnitude than the effect of having attended school, and almost as large as the difference in the total fertility rate of SSA and other low-income regions.

Table 1: Spouses' beliefs in ancestors and female fertility in Benin

	Total number of children ever born							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.750***	0.641***	0.629***	0.784***	0.825***	0.696***	0.701***	0.737***
	(0.178)	(0.161)	(0.158)	(0.195)	(0.198)	(0.180)	(0.177)	(0.216)
Baseline controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended controls	No	No	Yes	Yes	No	No	Yes	Yes
Religion FE	No	No	No	No	Yes	Yes	Yes	Yes
1 km buffer FE	No	No	No	Yes	No	No	No	Yes
Mean Y	5.365	5.352	5.352	5.398	5.365	5.352	5.352	5.398
R-Squared Y	0.0206	0.250	0.271	0.451	0.0534	0.272	0.292	0.466
N	846	840	840	640	846	840	840	640
Altonji et al. (2005) ratio:							14.43	58.29
Oster (2019): $R_{max} = 1.3 \times R'$ and δ=1							$\delta^* = 59.15$	$\delta^* = 244.02$

Note. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Columns 4 and 8 restrict the sample to households that have at least one neighbor within 1km from them, and include 1km-buffer fixed effects. The Altonji et al. (2005) ratio and Oster (2019)'s test use the coefficient from column 1 and those from column 7 or 8. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

[35]This specification also helps to rule out that the results are driven by the spatial clustering of groups with different fertility rates.

Robustness.- Appendix G provides several robustness tests. I show that the results are robust to estimating an exponential regression model using a pseudo-Poisson maximum likelihood estimator, or using alternative definitions of the explanatory and dependent variables.[36] I also show in specification curves that the coefficients barely move when including additional control variables,[37] that the results are not driven by similar dynamics in fertility and ancestral beliefs over the life cycle or over time (same results when including age fixed effects), or when using the total number of *alive* children as dependent variable, to reduce concerns about differential mortality. I also rule out that the effects are driven by differences in the inheritance rules of ethnic groups (Fontenay et al., 2024), by differences in old-age security that could be correlated with ancestral beliefs (Rossi and Rouanet, 2015; Rossi and Godard, 2022; Sage, 2023), or by the spatial clustering of groups with different fertility levels.[38]

4.2. Postcolonial Democratic Republic of the Congo

I then examine the same relationship using individual-level data from women born during the colonial period in the DRC coming from two original post-independence demographic surveys (EDOZA, which was conducted in both rural and urban areas, and an urban survey, conducted in 7 large cities of the DRC). Using self-reported ethnicity, I match women in these surveys to the information on the practice of ancestor worship before European colonization from Vansina (1966)'s book. I estimate a model similar to equation 1, but the explanatory variable of interest is now measured at the ethnic group level.

[36] Alternative measures of beliefs in ancestors are: dummy variable that equals one if the respondent answers *yes* to the question *Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?*, or when I combine both measures and generate a third variable that equals one when at least one of the two variables takes the value one.

[37] These variables include a dummy variable that takes the value 1 if respondent have experienced pressure from their extended family or clan to increase their number of children, polygamy, overall religiosity (measured as an indicator if the respondent donates money to the church when COVID happened), household size, or having livestock.

[38] An additional concern in this setting could be reverse causality. In fact, having more kids could reinforce people's thinking about lineage and heritage, and make people feel blessed and thankful to ancestors. However, I think that this is unlikely to explain my results for several reasons. First, beliefs in ancestors are deeply-rooted cultural beliefs, whose origin is uncertain, and unlikely to emerge at the individual level. Second, the results with the historical data from the DRC alleviate this concern (see Section 4.2), since ancestral beliefs are measured at the ethnic group level, using information on the practice of ancestor worship before colonization from Vansina (1966).

The nature of the 1970s Demographic Urban sample, composed mainly of post-independence migrants from rural areas, allows me to include city of residence fixed effects in the analysis. Therefore, in the spirit of an epidemiological approach, I identify the influence of ancestral beliefs by comparing people who lived in the same city at the time of the survey, but who migrated from different areas and have different beliefs in ancestors. This strategy is useful to rule out confounding factors related to the environment in which individuals live in (either local economic conditions, institutions, or supply-side frictions). For a formal exposition of the regression model see Appendix H.

Table 2 shows the results. Columns 1, 2 and 3 use both surveys together, while columns 4, 5 and 6 only use EDOZA (survey conducted in four regions in the west of the DRC, including both rural and urban areas), and columns 7, 8 and 9 only use the urban sample. In order to approximate complete fertility (these demographic surveys have information on all women over 13, and 75% of women under 20 are childless) and increase comparability with the first-hand data from Benin, the sample in columns 1, 2, 4, 5, 7 and 8 is restricted to women over the age of 30, while in columns 3, 6 and 9 it is restricted to women over 40.

Belonging to an ethnic group with ancestral beliefs is associated with about one additional child (20% of the outcome mean in the full sample). The difference between urban and rural areas is not very large: the effect of ancestor worship on fertility in the EDOZA sample (that includes rural areas) of women over 30 is 22% of the mean, while it is 16% in the urban sample (note that when using the urban sample, I use the city of residence FE instead of the region of residence FE).[39] Finally, I report in the last two rows of the table the Altonji et al. (2005)'s ratio and Oster (2019)'s test, which suggest that unobserved factors are very unlikely to fully explain the results.

[39] Interestingly, the fertility rates of women over 30 are higher in the urban sample than in EDOZA. This is consistent with previous work examining fertility rates in large cities such as Kinshasa (Shapiro, 1996).

Table 2: Ancestral beliefs and fertility in the DRC

	Total number of children ever born								
	Full sample			EDOZA			Urban sample		
	+30		+40	+30		+40	+30		+40
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Ancestral beliefs	1.082***	0.861***	1.140***	1.325**	1.052***	1.344***	0.836***	0.833***	1.189***
	(0.283)	(0.205)	(0.265)	(0.525)	(0.387)	(0.401)	(0.213)	(0.173)	(0.291)
Controls	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
City fixed effects	No	No	No	No	No	No	Yes	Yes	Yes
Mean Y	5.322	5.322	5.080	4.867	4.865	4.693	5.542	5.542	5.359
R-Squared	0.0799	0.246	0.243	0.108	0.267	0.273	0.0651	0.252	0.248
N	23212	23194	12435	7558	7551	5207	15654	15643	7228
Altonji et al. (2005) ratio:		1.17			1.13			2.45	
Oster (2019): $R_{max} = 1.3 \times R'$ and δ=1		$\delta^* = 4.35$			$\delta^* = 3.72$			$\delta^* = 10.09$	

Note. Data: Urban Demographic Survey of 1970s and EDOZA (rural and urban areas). Columns 1, 2 and 3 combine both the Urban Demographic survey and EDOZA. Columns 4, 5 and 6 use only the EDOZA sample. Columns 7, 8 and 9 use only the urban sample. In columns 1, 2, 4, 5, 7 and 8 is restricted to women older than 30 years old, while in columns 3, 6, and 9, the sample is restricted to women older than 40 years old. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the ethnic group e practiced ancestor worship before European colonization. Columns 1, 4 and 7 only include age and region fixed effects as controls. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City fixed effects are included for the urban sample and region fixed effects included in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Robustness.- Appendix H shows that these results are not affected by the sample restrictions (using age-specific fertility rates instead of total number of children ever born, or looking at the extensive vs the intensive margin), by important confounders that have been shown to affect fertility such as unobserved differences in human capital (Fernández and Fogli, 2009) or exposure to missionaries (Guirkinger and Villar, 2022), or are robust to the use of pseudo-Poisson maximum likelihood (since the number of 0's in the dependent variable is non-negligible).

4.3. *External validity: ancestral beliefs and fertility in different contexts and time periods*

The results presented above are not a specificity of the Beninese and Congolese contexts. In this section, I leverage additional large-scale datasets that cover different time periods and contexts and replicate the above findings.

Evidence from 19 contemporary Sub-Saharan African countries.– First, I look at the relationship between ancestral beliefs and fertility using individual-level data that covers more than 20.000 respondents from 19 sub-Saharan African countries (Pew Research

Center, 2009). With these data, I exploit the within-country variation in ancestral beliefs measured at both the individual and the regional level.[40] Consistent with previous results, I find a positive association between ancestral beliefs and fertility across SSA, ranging from about 10% of the sample mean (individual measure) to about 16% (regional measure). Interestingly, these data allow me to rule out overall religiosity or alternative supernatural beliefs as confounders in the relationship between ancestral beliefs and fertility. See Appendix I.1 for details about the regression model, results, and robustness tests.

Ancestral beliefs and fertility in traditional Folklore.– Second, I investigate whether the positive correlation showed above is detectable in the oral traditions of ethnic groups, leveraging information from the folklore catalog constructed by Michalopoulos and Xue (2021).[41] I find a positive and strong association between the share of motifs in the oral tradition of an ethnic group related to *fertility* or *birth* and the share of motifs related to *ancestors*, both at the global level and within Sub-Saharan Africa.[42] Appendix I.2 provides details on the regression model, results, and robustness tests.

Contraceptive use.– A natural implication of the results and hypotheses presented above is that contraceptive uptake, conditional on supply frictions, might be lower in societies with ancestral beliefs. I provide some results in this direction in Appendix I.3 using data from all available rounds of the DHS for Benin,[43] where there is significant variation in traditional beliefs at the individual level and adherence to African Tradi-

[40] Ancestral Beliefs are measured, at the individual level, as a dummy variable equal to one if the respondent answers "*Yes*" to the question "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*". At the regional level, they are measured as the prevalence of ancestral beliefs in each respondents region of residence (measured as the fraction of survey participants in a given region who answer "yes" to the aforementioned question).

[41] Figure D2 in Appendix D.2 plots the distribution of the share of motifs in an ethnic group's oral tradition related to ancestors. As it is clearly visible, the presence of ancestors in the folklore of ethnic groups is not specific of sub-Saharan Africa. We find extensive presence of ancestors in the oral tradition of societies in America and South Asia and the Pacific.

[42] One concern may be related to whether oral traditions are a good proxy of the actual relevance of these concepts. Here, I assume that, if the share of motifs related to ancestors in an ethnic group's oral tradition is large, then the beliefs in ancestors are likely to be strong. Although beliefs in ancestors and fertility change over time and therefore these measures are difficult to validate, Michalopoulos and Xue (2021) show that episodes in folklore accurately reflect the physical environment of ethnic groups (i.e., proximity to earthquake zones or the intensity of lightning strikes), which increases my confidence in using the motifs as proxies for the importance of fertility and ancestor worship.

[43] I also report results using the DRC DHS data, matched with the information on ancestral beliefs from Vansina (1966). Unfortunately, since both datasets are matched based on location, I cannot include place of residence fixed effects to account for supply-side factors.

tional religion (Voodoo) is a good proxy for traditional beliefs.[44] I find that, conditional on supply-side factors (including place of residence fixed-effects) and other important determinants of contraceptive uptake (including age, education, wealth, employment, marital status, or polygamy), traditional beliefs are associated with lower contraceptive use and intention to use. These results may help us to better understand the low demand for contraceptives in the context of SSA, where it has been shown that financial constraints or limited access to contraceptives are sometimes not important drivers of fertility rates (Dupas et al., 2024), and that fear to infertility is an important barrier (Ochako et al., 2015; Bau et al., 2024).[45]

5. Conceptual Framework: Mechanisms

In this section, I develop a very simple model of fertility where the total number of children in the lineage is a public good that highlights the relevance of continuing one's lineage in societies with ancestral beliefs. Moreover, this model provides very specific predictions that, if validated in the data, go in line with the motive to continue the family line driving fertility upwards, and are very difficult to explain by alternative explanations. In a context where getting exogenous variation in religious and cultural beliefs is extremely challenging, the test of the model's predictions in the data provides useful insights into the validity of the hypothesis at hand.

In the model, the agent derives utility from three components: consumption (c_i), the net monetary benefit of having children (n_i), and the total number of children belonging to the lineage (N).[46] Additive separability is assumed to represent preferences:

$$U_i = u(c_i) + b(n_i) + \beta\ell(N)$$

Where the functions $u(.)$, $b(.)$, and $\ell(.)$ satisfy the standard assumptions that $u' > 0$, $b' > 0$, $\ell' > 0$, $u'' < 0$, $b'' < 0$, $\ell'' < 0$. The extra term (ℓ) represents the utility that the

[44]Voodoo is explicitly mentioned in the constitution as an official religion, acquiring the same status as "missionary religions". Since Voodoo is not socially nor politically marginalized, concerns about undereporting of traditional religion in this context are limited.

[45]Fear to infertility may be particularly high in societies with ancestral beliefs since low fertility or infertility are associated with misfortune, sin, mistreatment, or marginalization, showing the disapproval of ancestors.

[46]Note that I do not need to assume the existence of a patriarchal household where the couple head has the final say in fertility decisions, as it is sometimes assumed in the context of SSA (Dupas et al., 2023).

individual derives from extending his/her lineage into the future, or in other words, it represents the motive to continue the family line. The parameter β takes the value 0 if ancestral beliefs are absent in the society and 1 if ancestral beliefs are prevalent.[47]

The composition of the total number of children belonging to the lineage (N) depends on the individual's gender and on the structure of the kinship system.[48] In patrilineal societies, both spouses and their children belong to the husband's lineage, as do the children of the husband's male siblings. On the other hand, in matrilineal societies, the husband does not belong to the lineage of his wife, and the children belong exclusively to the mother's line. Moreover, the children of one's sisters always continue one's lineage. Therefore, the total number of children belonging to the individual's lineage is different depending on the individual's gender and kinship system:

$$\text{For men:} \quad N = \begin{cases} n_i + \sum_{j \neq i} n_j & \text{if patrilineal, j indexes husband's brothers} \\ \sum_{k \neq i} n_k & \text{if matrilineal, k indexes own sisters} \end{cases}$$

$$\text{For women:} \quad N = \begin{cases} n_i + \sum_{j \neq i} n_j & \text{if patrilineal, j indexes husband's brothers} \\ n_i + \sum_{k \neq i} n_k & \text{if matrilineal, k indexes own sisters} \end{cases}$$

As we can see from above, the maximization problem is the same for both spouses in patrilineal societies, while it differs in matrilineal societies. In other words, preferences are aligned in patrilineal societies, where both spouses belong to the same lineage. In contrast, the motive to continue the family line diverges in matrilineal societies, since the husband does not belong to his wife's lineage and his family line cannot be continued by having children. I start by describing the problem faced by the agent in the patrilineal case, that it is the same for both spouses, and then will compare it to the problem in the matrilineal case, distinguishing between the husband and the wife perspectives.

[47]The parameter β could take values between 0 and 1 and be interpreted as a factor representing the intensity of ancestral beliefs. For example, it could be the subjective probability that the ancestors will be reincarnated in each of the new children belonging to the lineage.

[48]See Appendix L for a simple introduction to kinship systems.

5.1. *Patrilineal case*

I first study the solution under patrilineality. In this case, the individual's lineage is expanded through his/her own children as well as through the children of the husband's brothers. Both spouses therefore face the same maximization problem:

$$\max_{c_i,n_i} \; U_{\rm i} = u(c_i) + b(n_i) + \beta\ell(n_i + \sum_{j\neq i} n_j)$$

$$\text{s.t.} \;\; c_i + n_i \leq y_i$$

The first-order conditions for an interior solution give the following optimality condition:

$$u'(c_i) = b'(n_i) + \beta\ell'(n_i + \sum_{j\neq i} n_j) \tag{2}$$

Note that since the term $\beta\ell'(n_i + \sum_{j\neq i} n_j)$ is always positive, the optimal number of children is higher than in the absence of ancestral beliefs. Condition (2) tells us that an individual's utility depends not only on one's number of children, but also on the total number of children of the husband's brothers (i.e., the number of newborns who continue the lineage). Therefore, we need to better understand the interaction between the two. We can define the best response function as:

$$r_i(n_j) = \text{argmax}_{n_i} U_i\,(c_i, n_i, N)$$

Substituting it in the optimality condition of equation 2, we have that for interior solutions:

$$F = u'(c_i) - b'(r_i(n_j)) - \beta\ell'(r_i(n_j) + \sum_{j\neq i} n_j) = 0$$

Using the implicit function theorem:

$$\frac{dr_i\,(n_j)}{dn_j} = -\frac{\frac{\partial F}{\partial n_j}}{\frac{\partial F}{\partial r_i(n_j)}} = -\frac{\frac{\partial^2 U_i(c_i,n_i,ni+\sum n_j)}{\partial ni \partial n_j}}{\frac{\partial^2 U_i(c_i,n_i,ni+\sum n_j)}{\partial n_i^2}} < 0 \tag{3}$$

Expression 3 implies that when both n_i and n_j are positive, an increase in one leads to a decrease in the other, to maintain the equilibrium between the marginal utility of consumption and children. This indicates a negative relationship between an individuals number of children and that of her husbands brothers. The intuition is that, when the husbands brothers have many children, the individual has less incentives to contribute to the continuation of the family line, since the children of the husband's brothers already do so. Conversely, when their fertility is low, the agent increases her own to help sustain the lineage. This dynamic is akin to a classic free-rider problem, in which individuals benefit from others contributions to a public good while limiting their own.[49]

5.2. *Matrilineal case*

Let's turn now to the matrilineal case. In matrilineal societies, spouses have different incentives, since children only belong to their mother's family line.

5.2.1. *Men*

In matrilineal societies, the man's lineage is only continued by the children of his sisters, and therefore the maximization problem becomes:

$$\max_{c_i,n_i} \; U_\text{i} = u(c_i) + b(n_i) + \beta\ell(\sum_{k\neq i} n_k)$$

$$\text{s.t.} \;\; c_i + n_i \leq y_i$$

The first-order conditions for an interior solution gives the following optimality condition:

$$\frac{\partial U_i(c_i, n_i, N)}{\partial c_i} = u'(c_i) = b'(n_i) = \frac{\partial U_i(c_i, n_i, N)}{\partial n_i} \tag{4}$$

This condition states that the marginal utility of private consumption must equal the marginal utility of having children. As compared to expression 2, the term associated with the motive to continue the family line has dissapeared. This is because, in matrilineal societies, children do not belong to the lineage of their father and, as a result, the desire to continue the family line does not influence their fertility decisions. Finally,

[49]Note that this result cannot be fully explained from an evolutionary perspective. In evolutionary terms, inclusive fitness (the ability of an individual to pass on its genes-both through its offspring and through the offspring of close relatives with whom it shares genes) is increased when siblings have children, and thus the pressure to do so from one's own offspring is reduced. The difference with an evolutionary explanation is that it is not just the total number of siblings that matters, but also their sex composition. In patrilineal societies with ancestor worship, we expect a negative relationship between the number of *male* siblings and one's own fertility.

comparing equation 4 with the patrilineal case (equation 2), we see that men's average fertility is lower under matrilineal systems.

5.2.2. *Women*

From the woman's point of view, the optimization problem is slightly different than the man's. Now, the total number of children belonging to the woman's lineage is composed of both her own children and her sisters' children.

$$
\begin{array}{ll}
\max_{c_i, n_i} & U_{\mathrm{i}} = u\left(c_i\right) + b\left(n_i\right) + \beta \ell (n_i + \sum_{k \neq i} n_k) \\
\text{s.t.} & c_i + n_i \leq y_i
\end{array}
$$

From the first order conditions follow that:

$$
u'(c_i) = b'(n_i) + \beta \ell'(n_i + \sum_{j \neq i} n_k)
$$

As in the patrilineal case, the study of the interaction between n_i and n_k gives us the following free-riding condition:

$$
\frac{dr_i\left(n_k\right)}{dn_k} = -\frac{\frac{\partial F}{\partial n_k}}{\frac{\partial F}{\partial r_i(n_k)}} = -\frac{\frac{\partial^2 U_i(c_i, n_i, ni + \sum n_k)}{\partial ni \partial n_k}}{\frac{\partial^2 U_i(c_i, n_i, ni + \sum n_k)}{\partial n_i^2}} < 0 \tag{5}
$$

Condition 5 tells us that, in the case of matrilineal societies, there is a negative relationship between *female* fertility and the number of children of the woman's sisters, since they are strategic substitutes as both continue the woman's family line. Interestingly, when we compare the woman's optimality condition in the matrilineal case with the optimality condition in the patrilineal case, we do not observe important differences, as it was the case for men.

The model therefore suggests that, in societies with ancestral beliefs, female fertility levels should not be very different across kinship systems, whereas men want more children in patrilineal as compared to matrilineal societies. Note that this conclusion holds without imposing any particular bargaining weights between spouses, simply because preferences are aligned in patrilineal societies, but diverge in matrilineal societies. The same conclusions will therefore be reached if we assume that the male household head is the (almost) sole decision-maker than if we assume that both spouses have equal decision-making power. This simple model is therefore consistent with the following predictions:

- *Proposition 1*: On average, fertility is higher in societies with ancestral beliefs.
- *Proposition 2*: The positive influence of ancestral beliefs on fertility should be stronger in patrilineal ethnic groups, where both parents and the children belong to the same lineage.
- *Proposition 3*: In patrilineal societies with beliefs in ancestors, there is a negative relationship between one's own fertility and the number of male siblings' children.
- *Proposition 4*: In matrilineal societies with beliefs in ancestors, there is a negative relationship between female fertility and the number of women sisters' children.

In the next section, I provide empirical evidence to support these predictions.

6. Test of the theory

Having shown that ancestral beliefs are positively correlated with fertility levels, I now turn to empirically examine whether this relationship is driven by the main mechanism suggested in the qualitative literature: the stronger motive to continue the family line in societies with ancestral beliefs. Although the results shown in Section 4 hold in different samples and time periods, and are robust to the inclusion of many different control variables, the positive association between ancestral beliefs and fertility could still be capturing unobserved confounders.

In this section, I show that these results are specifically due to the motive to continue one's lineage. To do this, I test the very specific predictions provided by the simple model of fertility decisions of Section 5. The validation of these predictions in the data is consistent with a setting where the motive to continue the family line drives fertility upwards, and are difficult to explain by alternative hypotheses.

6.1. Kinship structure, ancestral beliefs and fertility behavior

I start by investigating the relationship between ancestral beliefs and fertility by kinship system. I test the second prediction of the model, namely that the positive influence of ancestral beliefs on fertility is driven by patrilineal ethnic groups, in two different settings. First, focusing on the Democratic Republic of Congo, I combine the ethnicity-level information on the practice of ancestor worship from Vansina (1966) with the type of kinship system from Murdock (1967), and match this information to the DRC DHS clusters using ethnic group boundaries.[50] About half of the sample lives in a DHS cluster belonging to the ancestral territory of a patrilineal ethnic group, and 13% live in clusters belonging to the ancestral territory of an ethnic group that did not practice ancestor worship. In all specifications, I include a vector of individual-level control variables such as age, age squared, gender, whether the respondent is Catholic, single years of education, whether the respondent lives in a urban or rural area, whether the respondent belongs to the top 40% of the wealth distribution, and dummies for provinces. Finally, since the variable measuring ancestral beliefs is defined at the DHS cluster level, I cluster the standard errors at that level. Second, I use ethnicity-level information at a global scale from Berezkin (2015)s catalog and the Ethnographic Atlas (Murdock, 1967).

Tables 3 and 4 reports the results. Consistent with the theoretical framework, I find that, in both cases, the positive influence of ancestral beliefs on fertility is driven by patrilineal ethnic groups, even at the global level during precolonial times. In the DRC, individuals from ethnic groups who practiced ancestor worship have higher fertility levels, larger ideal family sizes, and want more children in the future, but only if they belong to a patrilineal ethnic group. Interestingly, spouses' preferences are substantially affected (column 4 in table 3). In patrilineal societies with strong beliefs in ancestors, husbands want more children than their wives. However, in matrilineal societies with beliefs in ancestors, this relationship is reversed, and now wives want more children than their husbands. This is because in matrilineal societies children only belong to their mother's line, and therefore the motive to continue one's lineage is absent for men. Overall, these results go in line with the first prediction of the simple framework depicted in Section 5 – that is, the relationship between ancestral beliefs and fertility is stronger in patrilineal societies, where both parents and the children belong to the same lineage.

[50]The matching procedure between DHS clusters and Vansina (1966)' information is described in Section 3.

Table 3: Ancestral beliefs, fertility, and kinship structure

	(1) Nb of children	(2) Ideal Nb of children	(3) Want more children	(4) Husband > wife
Ancestor worship	-0.165	-0.0988	-0.0346	-0.102**
	(0.104)	(0.228)	(0.0268)	(0.0503)
Ancestor worship x Patrilineal	0.263**	0.422*	0.0593**	0.144**
	(0.126)	(0.254)	(0.0293)	(0.0561)
Province FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Mean Y	3.086	6.445	0.760	0.474
R-squared	0.622	0.241	0.221	0.0631
N	35891	33188	29568	10793

Note. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The sample is restricted to women in column 4. The outcome variable in column (1) is the total number of children ever born. In column (2), it is the ideal number of children. In column (3) it is a dummy variable that equals one if the respondent wants to have an additional child. Finally, in column (4), a dummy variable that equals one if the husband wants more children than his wife. "Ancestor Worship" is a dummy variable that equals one if the DHS cluster c of individual i belongs to the ethnic homeland of an ethnic group that traditionally practice ancestor worship as reported in Vansina (1966). "Patrilineality " is a dummy variable that equals one if the DHS cluster c of individual i belongs to the ethnic homeland of a patrilineal ethnic group as reported in Murdock (1967). Both interaction terms are included as controls. Controls also include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table 4: Ancestral beliefs, fertility and kinship structure in Folklore

	Share of motifs related to *birth*				Share of motifs related to *fertility*			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.00983	0.00648	0.00182	-0.0143	0.0160	-0.00549	-0.00815	-0.00188
	(0.0150)	(0.0160)	(0.0204)	(0.0265)	(0.0318)	(0.0311)	(0.0349)	(0.0400)
Ancestral beliefs x Patrilineal	0.0430***	0.0441***	0.0498**	0.0649**	0.124***	0.131***	0.124***	0.121***
	(0.0151)	(0.0153)	(0.0200)	(0.0259)	(0.0320)	(0.0315)	(0.0355)	(0.0406)
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	4.102	4.102	3.877	3.848	0.736	0.736	0.687	0.660
N	1228	1228	951	862	1228	1228	951	862

Note. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group, while in columns (5)-(9), it is the share of motifs related to "fertility". The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. "Ancestral beliefs" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. "Patrilineality" equals one if the major descent type of the ethnic group is patrilineal descent. Both interaction terms are included as controls. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Discussion and robustness.- One potential concern when interpreting the results showed above is omitted variable bias related to differences between patrilineal and matrilineal societies. However, that unobserved differences between patrilineal and matrilineal societies drive the results is unlikely for several reasons. First, I control for an extensive set of ethnic-level controls when using the Folklore data, which already reduces omitted variable concerns.[51] Second, although including these additional controls in the DRC sample reduces sample size and removes almost all the variation in ancestral beliefs, I show in Appendix J (table J1) that there are not significant differences between patrilineal and matrilineal societies in the DRC sample across important ethnic-level characteristics. Finally, I show in Appendix J (table J2) that the positive relationship between patrilineality and fertility only exists in societies with strong ancestral beliefs.[52] In fact, if the results were driven by inherent differences between patrilineal and matrilineal societies (e.g., in women's decision-making and empowerment in favor of women in matrilineal societies (Tene, 2020; Lowes, 2022), or to higher economic returns of children and the need of having sons for old age support (Rossi and Godard, 2022)), the positive association of patrilineality and fertility should be independent on ancestral beliefs.

6.2. Free-rider behavior

The third and fourth predictions generated by the model state that, in societies with strong beliefs in ancestors, there could be a negative relationship between one's own fertility and the number of family members able to continue one's lineage with their own children.[53] [54] In this section, I use data from the DRC DHS, combined with the in-

[51]These controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, the earliest year of publication in the group's oral tradition, whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, the share of motifs in an ethnic group related to "supernatural", tropical climate, rainfall, ruggedness, land quality (population weighted), and agricultural suitability.

[52]These results contribute to the literature on relationship between kinship structure and fertility (dating back to the pioneer work of Lorimer 1954). Moreover, I show that the positive relationship between fertility preferences and patrilineality in a context where the motive to continue the family line is important (such as in societies with strong ancestral beliefs) is particularly strong for men, as expected from the simple theoretical framework outlined in Section 5. See Appendix J for additional details.

[53]Strictly speaking, the model predicts a negative relationship between one's own number of children and the number of children of those in the family that belong to the same lineage. However, I look at the number of siblings instead since I do not have information about siblings' children in the DRC DHS data.

[54]This result helps to rationalize some empirical findings in other contexts where the motive to continue the family line is important, such as China. For example, Yang and Spencer (2022) find that "while the number of siblings of husbands and wives is of little or no consequence, the number of brothers of

formation on ancestor worship from Vansina's book and the Ethnographic Atlas, which contains information on the kinship structure of societies to examine this question. I proceed in two steps.

First, I focus on women and examine whether there exists a negative relationship between female fertility and the number of women's sisters in matrilineal societies with strong beliefs in ancestors. To do this, I rely on the information provided by the DHS on the gender and number of women's siblings. Second, I examine whether there exists a negative relationship between fertility and the number of eligible men (those aged 15-59) in the household in patrilineal societies with strong beliefs in ancestors.[55] [56]

Female fertility and the number of sisters.– Table 5 shows the results on the relationship between women's fertility and the number of sisters. Consistent with the intuition in section 5, we can see in column 1 (full sample) that, as compared to patrilineal societies, the number of sisters contribute negatively to fertility in matrilineal societies.[57] The fact that we already see a clear negative relationship in the full sample is explained by the fact that most of the sample belongs to societies with strong ancestral beliefs (87%). Moreover, all specifications control for the total number of alive siblings, in order to isolate the quasi-random variation in the gender composition of siblings and capture the effect of having more sisters rather than simply having more siblings in general.[58] Importantly, column 2 shows, in the spirit of a falsification test, that the number of brothers (who represent similar demographic variation but whose children do not continue

husbands matters: couples in which husbands have more brothers have fewer children. This is consistent with free-riding in a patrilineal kinship system, where fertility is at least partly driven by the motive to continue a family line.

[55]Unfortunately, the DHS does not include information on siblings for men. I use the number of eligible men (those aged 15-59) in the household as a proxy for the presence of family members capable of extending the family line.

[56]There are similar proportions of eligible men in the household by kinship system (52% of male respondents live in a household with at least one other eligible man in patrilineal societies, versus 45% in matrilineal societies). The main limitation of this analysis is that I only have information on the type of relationship to the household head for respondents, not for other household members, and therefore there may be compositional differences. For example, the probability that the other eligible male is a brother may be higher in patrilineal (and patrilocal) ethnic groups, and therefore my measure of eligible men in the household may be a better proxy for the number of male siblings in patrilineal ethnic groups.

[57]The fact that the coefficient in the pooled regression is more significant and a bit larger that the coefficients in columns 3 and 4 may be due to the larger number of observations used in the regression. When using information on ancestral beliefs from Vansina, I lose some observations since I am not able to match 100% of the DHS clusters to the ethnic homelands map in Vansina (1966).

[58]In contrast with much of Eastern Europe and Central and South Asia, there seems to be limited sex-selective abortions and stopping rule in Sub-Saharan Africa (Baland et al., 2023).

the lineage of the woman) has no significant relationship with fertility. This reinforces the interpretation that the negative influence of sisters in matrilineal societies captures the motive to continue the family line. An aspect of interest relates to the positive coefficients associated with the number of siblings, that may reflect the intergenerational transmission of preferences for larger families, or the support provided by the extended family to raise children.

Finally, columns 3 and 4 divide the sample for respondents who belong to an ancestral ethnic homeland with strong (column 3) or weak (column 4) ancestral beliefs. Consistent with the mechanism associated to the motive to continue the family line, the negative relationship between the number of sisters and fertility seems to be driven by societies with strong ancestral beliefs. Although the coefficient associated to *Matrilineal x N of Sisters* is 10 times higher in column 3, I am not able to reject the null of equality of coefficients due to the important differences in sample sizes (most of respondents belong to societies with strong ancestral beliefs), so these results should be interpreted with caution. Similarly, I show in column 5 that, when restricting the sample to societies with strong beliefs in ancestors, the negative relationship found in column 3 disappears when looking at the number of brothers.

Overall, these results provide suggestive evidence that the motive to continue the family line is at play in societies with strong ancestral beliefs, which drives fertility upwards as shown in the first part of the paper. Moreover, these findings improve our understanding of how people make fertility decisions within the larger framework of their families, clans, or lineages. Although it has long been recognized that fertility decisions are not driven solely by the interests of an individual (or couple), but rather are the result of a much broader collective interest (of the extended family or lineage) in the couple's fertility (see, for example, Smith 2004; Sear 2017), I uncover here a specific mechanism to substantiate these claims.

Table 5: Ancestral beliefs and free-riding behavior in matrilineal societies

	Total number of children ever born				
			Ancestors=1	Ancestors=0	Ancestors=1
	(1)	(2)	(3)	(4)	(5)
Number of sisters	0.0485***		0.0454***	0.0475*	
	(0.00986)		(0.0113)	(0.0268)	
Matrilineal x Nb of Sisters	-0.0308**		-0.0270*	-0.00254	
	(0.0142)		(0.0155)	(0.0543)	
Number of brothers		0.0357***			0.0386***
		(0.0101)			(0.0117)
Matrilineal x Nb of Brothers		-0.00933			-0.00582
		(0.0139)			(0.0155)
Province FE	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes
Mean Y	3.074	3.074	3.048	3.277	3.048
R-squared	0.614	0.614	0.627	0.550	0.627
N	26445	26445	21437	3233	21437

Note. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the total number of children even born. The sample is restricted to women. The sample is restricted to ethnic groups with ancestor worship in columns (3) and (5) and to ethnic groups without ancestor worship in column (4). "Matrilineal" is a dummy variable that equals one if the ethnic group e has matrilineal descent. "Number of sisters" is the number of sisters that the respondent has ever had. "Number of brothers" is the number of brothers that the respondent has ever had. The total number of alive siblings is always included as control variable. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Male fertility and the number of men able to continue the family line.– Although ideally one should explore the relationship between fertility and the number of the husband's brothers in patrilineal as compared to matrilineal societies, the DHS only includes information on siblings for women. Table 6 explores the relationship between one's fertility and the number of men aged 15-59 in the household, as a proxy for the number of husband's brothers. Consistent with the theoretical framework, the number of men in the household able to continue the lineage with their own children contributes negatively to fertility in patrilineal as compared to matrilineal societies (columns 1 and 2). Again, this negative relationship in the full sample is likely due to the high prevalence of societies with strong ancestral beliefs.

Dividing the sample by societies with strong and weak ancestral beliefs reveals that the negative effect of the number of eligible men on fertility is driven by societies with strong beliefs (column 3). In column 4, examining societies with weak ancestral beliefs shows that the negative relationship disappears in both statistical significance and magnitude. However, I cannot reject the null hypothesis of equality of coefficients; therefore, these results remain suggestive. Finally, the falsification test in column 5 shows that the negative pattern in column 3 between fertility and the number of men in patrilineal societies with strong ancestral beliefs does not extend to the number of women, suggesting that this negative relationship is related to the desire to continue the lineage rather than to the mere number of people living in the household.

Table 6: Ancestral beliefs and free-riding behavior in patrilineal societies

	Total number of children ever born				
			Ancestors=1	Ancestors=0	Ancestors=1
	(1)	(2)	(3)	(4)	(5)
Nb eligible men	-0.343***	-0.344***	-0.286***	-0.742***	
	(0.0423)	(0.0413)	(0.0421)	(0.191)	
Patrilineal x Nb of men	-0.105***	-0.0812**	-0.107***	0.0138	
	(0.0374)	(0.0376)	(0.0402)	(0.186)	
Nb eligible women					0.267***
					(0.0426)
Patrilineal x Nb of women					0.138***
					(0.0515)
Province FE	Yes	Yes	Yes	Yes	Yes
Basic controls	Yes	Yes	Yes	Yes	Yes
All controls	No	Yes	Yes	Yes	Yes
Mean Y	3.111	3.099	3.054	3.432	3.054
R-squared	0.640	0.647	0.659	0.596	0.659
N	12233	12049	9706	1515	9706

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the total number of children even born. The sample is restricted to men. The sample is restricted to ethnic groups with ancestral beliefs in columns 3 and 5, and to ethnic groups without ancestral beliefs in column 4. "Patrilineal" is a dummy variable that equals one if the ethnic group e has patrilineal descent. "Nb of eligible men" is the number of men aged 15-59 in the household. "Nb of eligible women" is the number of women aged 15-49 in the household. The total number of eligible people in the household is always included as a control variable. Basic controls include age, age squared and gender. All controls also include a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Validation across sub-Saharan Africa.– These results are not a specificity of the Congolese context. I show in Appendix J.1 that these results hold across sub-Saharan Africa by putting together 120 DHS surveys from almost 40 countries. First, since I do not have accurate information on beliefs in ancestors beyond the DRC and Benin, I look in table

J3 simply at the relationship between the siblings' gender and female fertility by kinship system.[59]

Second, I try to get closer to my preferred specifications and construct region-level measures of beliefs in ancestors using individual-level information from the Pew Research Center (2009). Then, I match these measures by location with the GPS location of DHS clusters, which leaves me with a (regional-level) measure of ancestral beliefs for about 800.000 women in 16 countries. With this information, table J4 in Appendix J.1 tests all model's predictions. Column 1 shows that ancestral beliefs correlate positively with fertility. Column 2 shows that this relationship is driven by patrilineal societies. Column 3 shows that, as compared to patrilineal societies, the number of sisters contribute negatively to female fertility in societies with ancestral beliefs.[60] Finally, column 4 includes a falsification test showing that the same negative relationship does not exist when looking instead at the number of the woman's brothers, since their children do not continue the woman's lineage in matrilineal societies.

6.3. *Additional implications: Ancestral beliefs and social organization*

As suggested by the results showed in the previous section, underlying the relationship between ancestors and fertility there is the more fundamental institution of lineage playing a pivotal role in these societies. In fact, ancestor worship belongs to the realm of kinship and lineage structures, since the whole system of beliefs, rituals and practices is centered around the clan or the extended family, and ultimately represents the extension of these aspects of the social organization to the supernatural sphere (Fortes, 1965). One interesting question then arises: will the traditional belief system of African societies, with a primary emphasis on the influence of ancestors on the everyday life of people, erode as the clan-based structure of societies dilutes?

I explore this question by making use of the distinction between lineage-based and age-based societies (Moscona and Seck, 2024).[61] As one of the primary goals of both

[59] Exploring these relationships is meaningful for the purpose of this paper if we assume that an important fraction of the population in sub-Saharan Africa holds ancestral beliefs and cares about the importance of continuing the family line. In this regard, these relationships are similar to the first column in tables 5 or 6.

[60] I report the results as a triple interaction since the regional-level measure of beliefs in ancestors is now continuous.

[61] In kin-based or lineage-based societies, the main social group is the extended family, and social loyalty

of living members and ancestors is the extension of the lineage into the future, I expect ancestor worship to play a weaker role in age-based societies (where "horizontal" organization is key) as compared to lineage-based societies (where "vertical" organization is key). I follow the classification of Moscona and Seck (2024) to identify in the contemporary PEW data which ethnic groups are organized along age lines and which ethnic groups are dominated by lineage organization.[62]

Table K1 in Appendix K shows the results. Although the estimates are less precise due to the reduced sample size, I find suggestive evidence that the positive influence of ancestral beliefs on fertility is stronger in kinship-based societies as compared to age-based societies.

7. Conclusion

Fertility is collapsing globally, in many places faster than expected, rising as one of the worlds most pressing problems and creating new challenges for economic growth and social stability (Bhattacharjee et al., 2024; Maestas et al., 2023). The limitations of existing theories in understanding fertility decline suggest room for complementary explanations that consider the importance and evolution of religious and cultural beliefs and practices. In this regard, the study of traditional religious and cultural beliefs in sub-Saharan Africa, the region with the highest fertility rates and slower fertility decline, which have been extremely understudied, may provide us with important insights to better understand fertility patters.

This paper examines the importance of deep-seated cultural norms, rooted in traditional beliefs, in explaining fertility rates in sub-Saharan Africa. I examine the role played

is within the kin group. In age-based societies, on the other hand, the main social group is the age group, i.e., a group of individuals of similar age who enter adulthood at the same time. Although ancestors are mainly concerned with the reproduction of the lineage, beliefs in ancestors also exist in age-set societies. First, because it is possible for an ethnic group to have both kinship and age-set structures. Although many groups have either a strong lineage structure or a strong age structure, there is a continuum of possibilities in between. Second, there may be different sets of ancestral spirits. For example, Kenyatta (1965) notes for the Gukuyu (an age-based society from central Kenya) that in addition to the spirits of the father and mother and the clan spirits, there are age-group spirits. The key argument here is that these spirits will play a different role in the society, and the motive of lineage continuation for fertility will be less important.

[62] Although they focus on both Kenya and Uganda, I focus only on respondents from Kenya because I have almost no variation in Uganda (96% of respondents from Uganda belong to an ethnic group where lineage structure dominates, versus 27% in Kenya).

by the belief in the influence of ancestors, a belief system that emphasizes ancestry and descent and the importance of continuing one's lineage (Lesthaeghe, 1989; Caldwell and Caldwell, 1987, 1988, 1990; Caldwell et al., 1992). I document and test the hypothesis that ancestral beliefs, which places great emphasis on the continuation of the family line and reinforces multigenerational family obligations, contributes to sustain high fertility rates. I find a positive association between beliefs in ancestors and fertility in different context and time periods, that holds across ethnic groups, individuals within regions and villages, and across migrants who live in the same city but come from places that differ in the practice of ancestor worship.

Then, I demonstrate how the positive correlation between ancestral beliefs and fertility is connected to the value placed on perpetuating one's lineage. To accomplish this, I test two main predictions derived from a simple model that captures the motive to continue the family line. First, I demonstrate that the influence of ancestral beliefs on fertility depends on the kinship system because the importance of this motive varies. I find that ancestral beliefs have a stronger positive effect on fertility in patrilineal societies, where both parents and children belong to the same lineage. In these societies, having children is always a way to continue one's family line. In contrast, in matrilineal societies, the father does not belong to his wife's lineage, and children belong exclusively to the mother's lineage. Thus, there is no motive for men to continue the family line with their own offspring.

Second, I identify specific free-rider behavior among siblings that depends on the kinship system. Specifically, I show a negative relationship between one's own fertility and the number of family members who can continue the family line with their own children. In matrilineal societies with strong ancestral beliefs, fertility decreases with the number of sisters (but not brothers), while in patrilineal societies with strong ancestral beliefs, fertility decreases with the number of men in the household who can continue the lineage (but not with the number of women).

These findings provide insights into the determinants of fertility by highlighting the potential role that culture and religious beliefs may play. In this regard, an interesting avenue for future research is to examine whether recent changes in the social fabric and culture of societies (e.g., the high and increasing individualism, the dominance of nuclear families, the absence of cooperative parenting, or the transformation of childbearing from a shared social duty to a personal choice) has affected the (faster than expected) fertility collapse in other parts of the world; or to examine whether factors that contribute to the dissolution of traditional beliefs systems such as beliefs in ancestors (e.g., new Christian "post-colonial" churches) will contribute to accelerate fertility decline in Sub-Saharan Africa.

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Online Appendix

Ancestral Beliefs and Fertility in Sub-Saharan Africa

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A. Prevalence of ancestral beliefs in SSA

In this section, I discuss why the belief in ancestors is more influential in sub-Saharan Africa than in any other region of the world. As noted by Caldwell and Caldwell (1987), ancestor worship is not historically unique to SSA, and may have been the original religion – understood here as a set of beliefs that structure the interaction between a community and the supernatural forces it perceives in the world – in many parts of the world, including Eurasia and the Americas.[63] Perhaps the most prominent and studied example outside SSA is South Asia, especially China, where it has been shown to influence fertility and son preference historically and even today (Ahern, 1973; Hu, 2016; Hu and Tian, 2018; Zhang, 2024).[64]

There are several factors that may explain the higher prevalence of beliefs in ancestors in sub-Saharan Africa compared to other regions of the world. First, the geographic isolation of SSA beyond the Sahara and the presence of several diseases prevented foreigners from gaining political power in the interior until the end of the 19^{th} century, limiting their presence to coastal trading posts and thus preventing the spread of alternative belief systems that emphasized the relationship between individuals and external gods and undermined kinship ties (Caldwell and Caldwell, 1987; Schulz et al., 2019; Schulz, 2022).

Second, the long tradition of high land-to-man ratios in SSA has prevented the erosion of lineage and clan structures in which ancestor worship is embedded (Boserup, 1965, 1985; Platteau, 2000). In fact, ancestor worship is the extension of these relations to the supernatural sphere (Fortes, 1965; Caldwell and Caldwell, 1985). Collective farms in land-abundant societies based on production activities such as shifting cultivation are efficiently managed by extended families or lineages. As population density increases and land abundance decreases, private property rights tend to emerge and family structures shift towards the conjugal family (Platteau, 2000; Guirkinger and Platteau, 2014).

[63]This hypothesis may be traced back to Victorian anthropologist Herbert Spencer, who believed ancestorism to be the root of every religion (Grande, 2024).

[64]Other ancient societies, such as the Greeks, did not develop ancestor worship but rather a cult of the dead. This distinction is important. For example, Catholic societies have a cult of the saints and organize masses for the dead, but they are not considered to be ancestor worshippers either (Fortes, 1965). The main difference between ancestor worship and the worship of the dead is the strong belief of the former in the influence of the ancestors in the affairs of the living.

Thus, agrarian societies in SSA are characterized by the presence of free peasants, free land, and the absence of non-working landowners, which reinforces lineage-based organization. For this reason, the *de facto* influence of beliefs in ancestors nowadays is stronger in sub-Saharan Africa than in regions where ancestor worship was traditionally widespread (e.g., Southeast Asia). As Platteau (2000) notes: "Asian societies have a strong tradition of settled agriculture, and property rights on arable lands have long been established with significant class differentiation or stratification in terms of land and other assets. Contrary to African societies which are centered around the lineage and are therefore permeated by values of extended solidarity, Asian societies, like all peasant societies (including European societies), have evolved over centuries social and family patters based on the conjugal family". As a result of low population density and the prevalence of lineage-based systems of social organization, the intensity of kinship ties and the emphasis on descent are higher in SSA than in any other region of the world, which has hindered the dissolution of ancestral beliefs, since ancestors often symbolize the continuation of a social structure centered on lineage and descent, and therefore organized hierarchically by age, with the oldest person often occupying the position of authority (Alidou and Verpoorten, 2019).[65]

Finally, historical and institutional factors specific to SSA have traditionally favored the reproduction and persistence of cultural traits over time (Platteau, 2009). The existence of strong kinship ties, in which ancestral beliefs are embedded, was facilitated by the existence of weak states and the need for protection against raids during the Arab slave trade (c. 950-1950), and reinforced by the impact of colonialism and indirect rule on the social and political structure of African societies. [66]

Because of these contextual barriers, the new religions that spread rapidly in the 20^{th}

[65]In fact, age (and gender) are the two main dimensions along which African societies are stratified. Again, this is closely related to the high land-to-man ratios in SSA. In contrast, Asian village communities have well-established property rights, and (scarce) land is more subject to market exchange, which tends to concentrate ownership in the hands of a class of landowners. Moreover, lineage organization and ancestral beliefs reinforce each other because elders are often considered closest to the ancestors, as in the case of the Bakongo in the DRC or the Kaguru in Tanzania. In addition, because the hierarchy does not end with death, ancestors acquire greater authority than any living person and thus the power to intervene in the lives and affairs of their descendants.

[66]The system of indirect rule reinforced the clan-based structure of societies and increased ethnic consciousness by benefiting from the creation of uniform groups composed of individuals identified as members of a particular ethnic group and placed under the control of an African official regarded as a tribal or village chief (Ekeh, 2004; Ellis and Ter Haar, 2004; Platteau, 2009).

century have had only a limited impact on African traditional supernatural beliefs (Platteau, 2009; Igboin, 2022). Instead, modern religions have adapted to coexist with traditional supernatural beliefs, combining faith in a High God with supernatural beliefs (Platteau, 2014). With respect to beliefs in ancestors, Igboin (2022) and McCall (1995) highlight that, although ancestors remain a central feature of many communities in Africa, there is little attention paid to ancestors and ancestor-related practices by scholars of African society and culture, in part due to the dominance of Western scholarship in this area of research.

For instance, data from the 2009 Pew data collection project on spiritual life (Pew Research Center, 2009), that contains information on traditional beliefs for around 25.000 respondents from 19 sub-Saharan African countries, confirms this intuition and shows that 45.6% of respondents believed in witchcraft, while 41% believed in reincarnation, and about 30% believed in ancestors.[67] In the case of my sample in Southern Benin, 50% of men think that their ancestors care about the continuation of the lineage (43% among those who do not report traditional religion as their main religion) or, in the same vein, Le Rossignol et al. (2022) find that in the northwestern DRC, 92% of their sample report believing either "strongly" or "very strongly" in Christianity, and at the same time, 75% of them also report believing "strongly" or "very strongly" in witchcraft.

B. Ancestors in the folklore and symbolism of communities

B.1. (Extract of) The Breath of the Ancestors (Birago Diop, 1960)

[67]Beliefs were recorded as yes or no answers to the question Which, if any, of the following do you believe in? and assigned to various categories such as witchcraft, the evil eye, evil spirits, heaven, or reincarnation. Ancestral beliefs were recorded as a yes or no answer to the question Do you believe that sacrifices to spirits or ancestors can protect you from bad things happening?.

B.2. English	*B.2. Original*
Listen more often	Écoute plus souvent
to things than to beings	les choses que les êtres
the voice of the fire	La voix du feu sétend
the water that speaks	entends la voix de leau
the voice of the wind	Écoute dans le vent
the bush that weeps	le buisson en sanglot

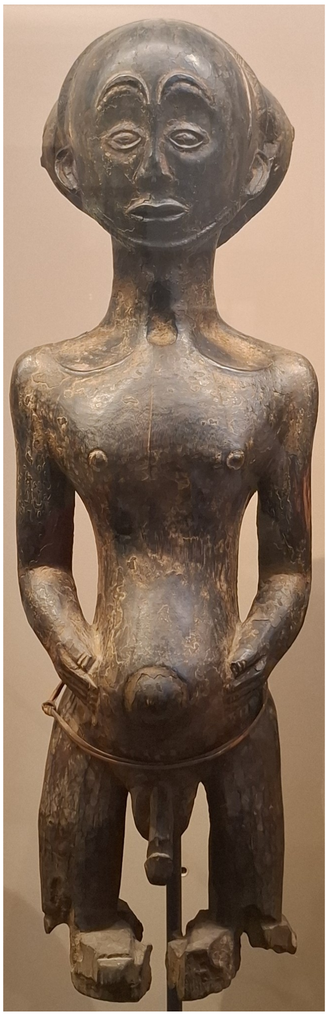
this is the ancestors breathing

The breath of the ancestors
who are not gone
who are not underground
who are not dead
those who have passed are not gone
they are in a womans breast
in a childs wailing song
in the coals that wont rest

Cest le souffle des ancêtres

Le souffle des ancêtres morts
qui ne sont pas partis
Qui ne sont pas sous terre
qui ne sont pas morts
Ceux qui sont morts ne sont jamais partis
Ils sont dans le sein de la femme
Ils sont dans lenfant qui vagit
Et dans le tison qui senflamme

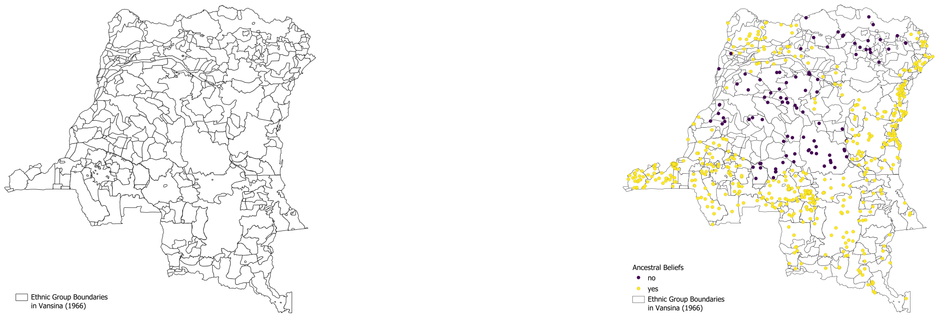
B.3. Ancestral statue, former Katanga province, RD Congo (20th century)



Source: Photo taken by the author. The statue is in the Royal Museum for Central Africa in Tervuren (Belgium).
From the description of the Royal Museum for Central Africa: This statue is the work of a Hemba artist. The hand on the stomach indicate the succession of generations.

C. Matching procedure: Vansina and DRC DHS

Figure C1: Ethnic groups in Vansina (1966) and DHS clusters



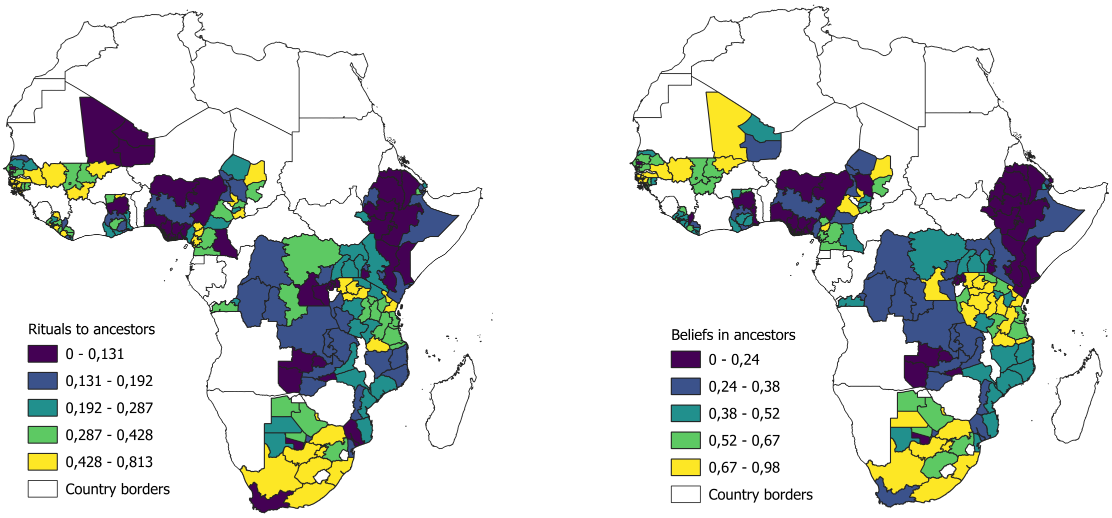
(a) Digitized version of Vansina (1966) (b) Ancestor worship and DHS clusters

Notes: Panel C1a displays ethnic group boundaries for the DRC digitized from Vansina (1966). Panel C1b assigns to each DHS cluster the value taken by the variable ancestor worship in the ethnic homeland where the DHS cluster is located.

D. Geographical distribution of variables: PEW and Folklore

D.1. PEW Variables

Figure D1: Share of respondents believing in Ancestors

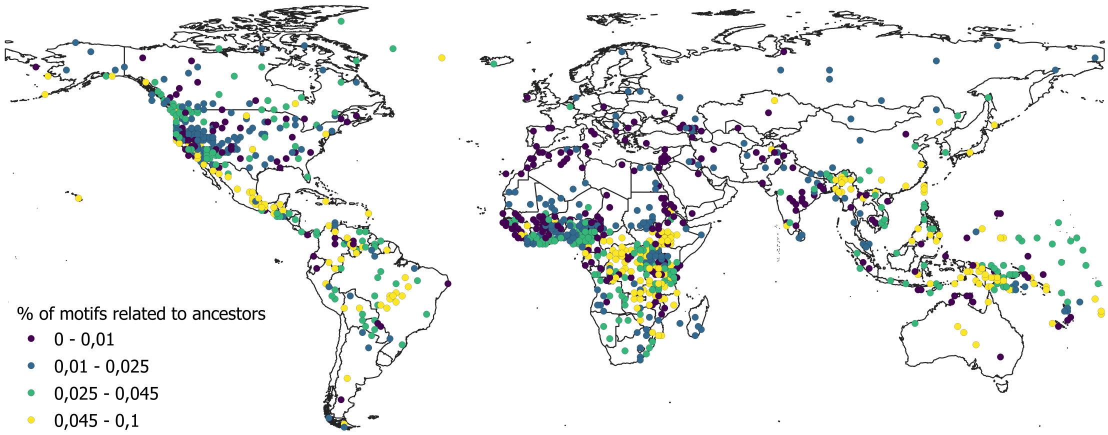


(a) Rituals to ancestors (b) Beliefs in ancestors

Notes: The figure displays the share of individuals in each administrative region of the PEW sample believing in ancestors. Panel D1a shows the geographic distribution of the share of respondents that answered "yes" to the question "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*". Panel D1b shows the geographic distribution of the share of respondents that answered "yes" to the previous question or to the question "*Do you believe that sacrificing to spirits or ancestors can protect you from bad things happening?*".

D.2. Distribution of motifs related to ancestors

Figure D2: Share of Motifs related to ancestors in the oral tradition of EA's ethnic groups



Note: The figure displays the location of the 1.265 groups in the Ethnographic Atlas (Murdock, 1967), and the distribution of the share of motifs related to the word "ancestor".

E. Variable definitions

E.1. Ethnicity-level variables

E.1.1. Folklore:

Number of motifs: Total number of motifs recorded in each Berezkin's group. Data comes from Michalopoulos and Xue (2021).

Number of publishers: Total number of publishers of the sources in the group's oral tradition. Data comes from Michalopoulos and Xue (2021).

Earliest year of publication: Earliest year of publication in the group's oral tradition. Data comes from Michalopoulos and Xue (2021).

Motifs related to "supernatural": Share of motifs tagged by supernatural related terms in the oral tradition of an ethnic group. Data comes from Michalopoulos and Xue (2021).

E.1.2. Ethnographic Atlas:

Domestic organization around extended families: is based on varible V8 of the Ethnographic Atlas (domestic organization). It is constructed as a dummy variable that is zero if domestic organization is "Independent polyandrous families", "Polygynous: unusual co-wives pattern", "Polygynous: usual co-wives pattern", "Minimal (stem) extended families", "Small extended families", and "Large extended families". It equals one if domestic organization is "Independent nuclear family, monogamous" or "Independent nuclear family, occasional polygyny". This measure is based on Enke (2019).

Segmented communities and localized clans: is based on variable V15 of the Ethnographic Atlas (community marriage organization). It is constructed as a dummy variable that is zero if community organization is "Demes, not segregated into clan barrios", "Agamous communities", "Exogamous communities, not clans". It takes on a value of one if community organization is "Segmented communities without local exogamy", "Segmented communities, localized clans, local exogamy", or "Clan communities, or clan barrios". This measure is based on Enke (2019).

Patrilineal: is based on variable V43 of the Ethnographic Atlas (descent: major type). It is a dummy variable that takes the value one if a groups major mode of descent is patrilineal as opposed to any other mode of descent.

Political complexity: is based on variable V33 of the Ethnographic Atlas (jurisdictional hierarchy beyond local communities). This is a categorical variable ranging from no levels beyond local community to four levels beyond local community. I use it as it appears in the Ethnographic Atlas (ranging from 0 to 4).

Monogamy: is based on variable V9 of the Ethnographic Atlas (marital composition: monogamy and polygamy). It is constructed as a dummy variable that takes on a value of one if an ethnic group's dominant form of marital composition is monogamous.

Pastoralism: is based on variables V4 (animal husbandry) and V40 (predominant type of animal husbandry) of the Ethnographic Atlas. To construct this variable, I follow Le Rossignol and Lowes (2022). From variable v4, I create a dummy variable that equals one if the predominant type of animal raised is a herding animal such as cattle, sheep, or camelids. Then, pastoralism is measured by multiplying this new dummy variable by variable V40.

Historical use of the plough: is based on variable V39 of the Ethnographic Atlas (animals and plow cultivation). I construct an indicator variable for traditional plough agriculture that equals one if the plough was present (whether aboriginal or not) and zero otherwise. This measure is similar to Alesina et al. (2013).

Historical economic development: is based on variable V30 of the Ethnographic Atlas (settlement patterns). I use this variable as in Alesina et al. (2013). Each ethnic group is categorized into one of the following categories describing their pattern of settlement: "nomadic or fully migratory", "semi-nomadic", "semi-sedentary", "compact but temporary settlements", "neighborhoods of dispersed family homes", "separated hamlets forming a single community", "compact and relatively permanent", "complex settlements". The variable takes on the values of 1 to 8, with 1 indicating fully nomadic groups and 8 groups with complex settlement.

Practice of intensive agriculture: is based on variable V28 of the Ethnographic Atlas (intensity of agriculture). I follow Alesina et al. (2013) and construct an indicator that equals one if the society belongs to the categories "intensive agriculture" or "intensive irrigated agriculture", and zero otherwise.

E.1.3. Geographic variables:

Tropical climate: is taken from Alesina et al. (2013). It is computed as the proportion of land within a 200 kilometer radius of an ethnic groups centroid that is classified as being either tropical or subtropical. The classification of thermal climates comes from the GAEZ 2002 database.

Precipitation: is taken from Fenske (2013). It is average annual precipitation (mm). The data comes from the International Institute for Applied Systems Analysis.

Ruggedness: is taken from Nunn and Puga (2012). It is constructed as the average across points on a grid 1 kilometer apart within a country of an index of terrain ruggedness. For details about how the index is constructed, see Nunn and Puga (2012).

Land quality: is taken from Fenske (2013). It is measured as a non-additive combination of climate constraints, soil constraints and terrain slope constraints, coming from the Food and Agriculture Organizations Global Agro-Ecological Zones (FAO-GAEZ) project.

Agricultural suitability: is taken from Alesina et al. (2013). Using information on global geo-climatic conditions for crop cultivation from the FAOs Global Agro-Ecological Zones (GAEZ) v3.0 database, they calculate the fraction of this land that is suitable for the cultivation of barley, wheat, rye, sorghum, foxtail millet, or pearl millet. They use this measure to construct the average suitability of the land (within 200 kilometers of the centroid of each ethnic group) historically inhabited by a locations ancestors.

F. West Zaire Surveys (1975-77)

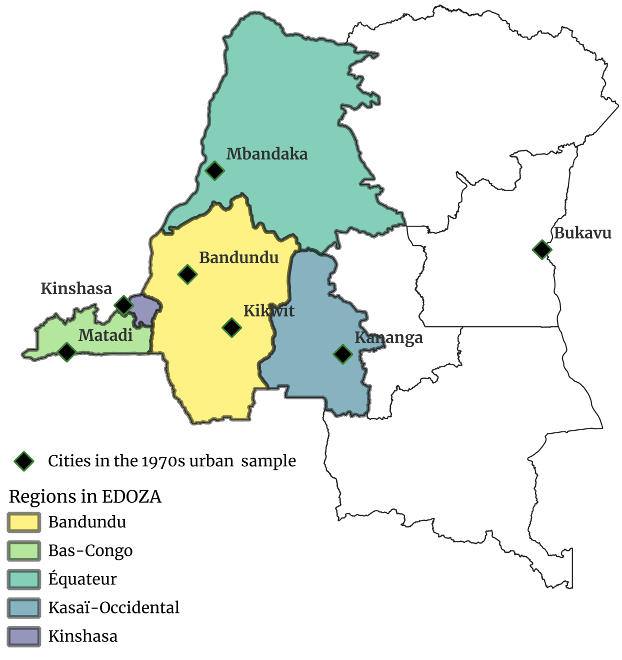
These surveys were part of a large research project led by the Université Catholique de Louvain in collaboration with the Congolese Official Institute of Research and Statistics. The surveys were conducted between 1975 and 1977 and aimed at collecting basic demographic information for the Democratic Republic of the Congo, due to the paucity of this type of data. These surveys are the outcome of two parallel studies (see Figure F1 for the sampled territories):

Urban Demographic and Budgetary Survey.–A demographic and budgetary survey were conducted in the six largest cities of western Zaire: Kinshasa, Matadi, Bandundu, Kikwit, Mbandaka and Kananga. This survey was led by Joseph Houyoux, professr at the Sociology department of the Université Catholique de Louvain, in Belgium. On the one hand, the demographic survey contains individual-level information on about 250,000 individuals in 43,000 households. After a census conducted in each of these cities to count and identify all households, 1/10 of the total number of households was randomly selected to be included in the sample. Demographic information was collected on age, sex, ethnicity, place of birth, and migration status, as well as socioeconomic characteristics such as years of education and occupation. Moreover, for each woman over the age of 13, births calendars are also recorded, including the month and year of each birth, the sex of the children, and the dates of death, if applicable. On the other hand, the budgetary survey contains information on household expenditures and transactions for 1/50 of the households identified in each city (two daily passages during one month).

Enquête Démographique de l'Ouest du Zaïre (EDOZA).–A demographic survey conducted in the rest of urban areas (excluding the large cities included in the previous survey) and in rural areas of four western regions: Bas-Congo, Bandundu, Kasai Occidental and Equateur (only in the Equateur and Tshuapa districts). This survey was led by the Demography deparment of the Université Catholique de Louvain. The original survey includes around 210.000 individuals in about 45.000 households. However, contrary to the Urban Demographic survey where I have the full sample, I have been able to locate 50.000 individuals in 11.000 households so far. To the extent of my knowledge, the rest of the EDOZA sample is lost.

One potential concern relates to the quality of these surveys. Although these surveys were conducted during Mobutu's dictatorship, both their design and implementation were managed jointly by a team of demographers based at the Congolese Institute for Research and Statistics and the Université Catholique de Louvain in Belgium. Data cleaning and statistical programming were carried out in Belgium. These data are considered high quality by demographers who rely on them to study demographic dynamics in the Congo in well-published academic papers (Tabutin, 1982; Shapiro, 1996; Shapiro and Tambashe, 2001; Schneidman, 1990). Finally, these surveys used the same methodology to construct variables (i.e., birth calendars) as current benchmark surveys, such as the Demographic and Health Surveys. Although the data from these surveys have been used by demographers, it has only been used in an aggregated way to study fertility and mortality levels and trends across regions and time. Recently, some papers in economics have used the microdata of the Urban Demographic survey (Guirkinger and Villar, 2022; Alvarez-Aragon et al., 2025). To the best of my knowledge, no paper has yet exploited the microdata from EDOZA.

Figure F1: Cities included in the urban survey and regions included in EDOZA



Note: Own elaboration using QGIS.

G. Ancestral beliefs and fertility in southern Benin

G.1. Effects on women and complementarity between spouses

Tables G1 (G2) presents the estimates of equation 1 for men (women) when using only the beliefs of the husband (wife) as explanatory variable. Finally, Table G3 shows the results of regressing male fertility on both husband and wife's beliefs. The same set of control variables as in Table G1 are included. The results show a systematic positive relationship between beliefs in ancestors and fertility, particularly strong when both spouses hold ancestral beliefs.

Table G1: Beliefs in ancestors and male fertility in contemporary Benin

	Number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	1.848***	1.667***	1.188***	1.460***	1.267***	1.152***
	(0.379)	(0.345)	(0.298)	(0.388)	(0.350)	(0.355)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	7.995	8.016	8.016	8.012	8.029	8.029
N	846	832	832	841	830	830

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table G2: Beliefs in ancestors and female fertility in Benin

	Number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.570***	0.423***	0.392***	0.630***	0.460***	0.430***
	(0.155)	(0.139)	(0.138)	(0.166)	(0.150)	(0.149)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	5.345	5.333	5.333	5.345	5.333	5.333
R-Squared Y	0.0139	0.237	0.256	0.0455	0.254	0.274
N	943	937	937	943	937	937

Note. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table G3: Spouses' beliefs in ancestors and male fertility in contemporary Benin

	Number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	2.695***	2.310***	2.066***	2.235***	1.825***	1.626***
	(0.519)	(0.478)	(0.473)	(0.533)	(0.493)	(0.488)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	7.995	8.016	8.016	8.012	8.029	8.029
N	846	832	832	841	830	830

Note. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

G.2. *Poisson model*

Table G4: Beliefs in ancestors and female fertility in Benin, Poisson regression

	Total number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.135***	0.114***	0.111***	0.149***	0.125***	0.125***
	(0.0310)	(0.0284)	(0.0278)	(0.0345)	(0.0317)	(0.0311)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	5.365	5.352	5.352	5.365	5.352	5.352
N	846	840	840	846	840	840

NOTE. Data: First-hand data from Benin. The sample is restricted to women. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the total number of children ever born. "Ancestral beliefs" equals one if both spouses answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?* Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, TV ownership, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

G.3. *Alternative explanatory variables*

Table G5: Beliefs in ancestors (alternative measure) and female fertility in Benin

	Total number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.486***	0.442***	0.423***	0.568***	0.504***	0.511***
	(0.160)	(0.142)	(0.139)	(0.173)	(0.155)	(0.152)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	5.365	5.352	5.352	5.365	5.352	5.352
R-Squared	0.0106	0.244	0.264	0.0444	0.266	0.287
N	846	840	840	846	840	840

NOTE. Data: First-hand data from Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" equals one if both spouses answer *yes* to the question "*Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?*" Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, TV ownership, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Table G6: Beliefs in ancestors (combined measure) and female fertility in Benin

	Total number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.486***	0.442***	0.411***	0.568***	0.504***	0.509***
	(0.160)	(0.142)	(0.140)	(0.173)	(0.155)	(0.152)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	5.365	5.352	5.352	5.365	5.352	5.352
R-Squared	0.0106	0.244	0.265	0.0444	0.266	0.287
N	846	840	840	846	840	840

Note. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" equals one if both spouses answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?* or to the question *Do you think that your ancestors' spirits care about your extended family and the continuation of the family line?* Baseline controls include age, age squared, electricity and education. Extended controls include the number of agricultural fields, TV ownership, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

G.4. *Number of alive children*

Table G7: Beliefs in ancestors and alive number of children in Benin

	Total number of alive children					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.475***	0.424***	0.410***	0.595***	0.525***	0.526***
	(0.149)	(0.142)	(0.141)	(0.157)	(0.150)	(0.149)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Mean Y	4.759	4.751	4.751	4.759	4.751	4.751
R-Squared Y	0.0122	0.204	0.206	0.0395	0.221	0.231
N	838	832	832	838	832	832

Note. Data: First-hand data from Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children alive at the time of the survey. "Ancestral beliefs" equals one if both spouses answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Voodoo, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

G.5. *Specification curves: sensitivity to additional controls*

In this section I show that the coefficients barely move when including alternative control variables and their different combinations. Under random assignment, point estimates should not dramatically change after the inclusion of control variables. In this sense, low sensitivity to the inclusion of control variables may reduce concerns about omitted variable bias. In addition to the baseline and extended controls (age, education, access to electricity, TV ownership, quantity of agricultural land, pressure from the extended family to have children and whether the respondent is participating in a program to increase pineapple production), I include:

Pressure from extended family to have children: To reduce concerns related to the fact that the measure of ancestral beliefs may be capturing the importance of extended family networks and the prevalence of clan-based relationships, I include a dummy variable that equals one if the respondent has experienced pressure from his/her extended family to have children.

Polygyny: Since I am focusing on married couples, polygyny is an important factor to consider. In fact, about 60% of men who report having 10 or more children are polygynous. I have not included polygyny in the main specification because it may be a "bad control" (Angrist and Pischke, 2009). For example, Goody (1973) notes that one can gain security in the afterlife (i.e., continuity of the family line) directly by having children or indirectly by adding wives. Similarly, Kenyatta (1965) mentions in the context of the Gikuyu people of central Kenya that:

> "If a man dies without a male child his family group comes to an end. This is one thing that the Gikuyu fear dreadfully, and it can be said to be one of the factors behind the polygamous system of marriage. There is no doubt that perpetuation of family or kinship group is the main principal of every Gikuyu marriage. For the extinction of a kinship groups means cutting off the ancestral spirits from visiting the earth, because there is no one left to communicate with them (Kenyatta (1965), p.13)".

Husband's father practices Vodoun: To proxy for parental preferences for children and aspects related to the traditional practices of the family, I include a dummy variable that equals one if the husband father's main religion is Voodoo. This variable is highly correlated to the respondent's religion: when the respondent declares Voodoo as his main religion, his father practices Voodoo in the 85% of cases.

Household size: This variable takes into account the structure of the household. Households with more members can more easily externalize the costs of having children, and more members may be the result of stronger preferences towards children in previous generations.

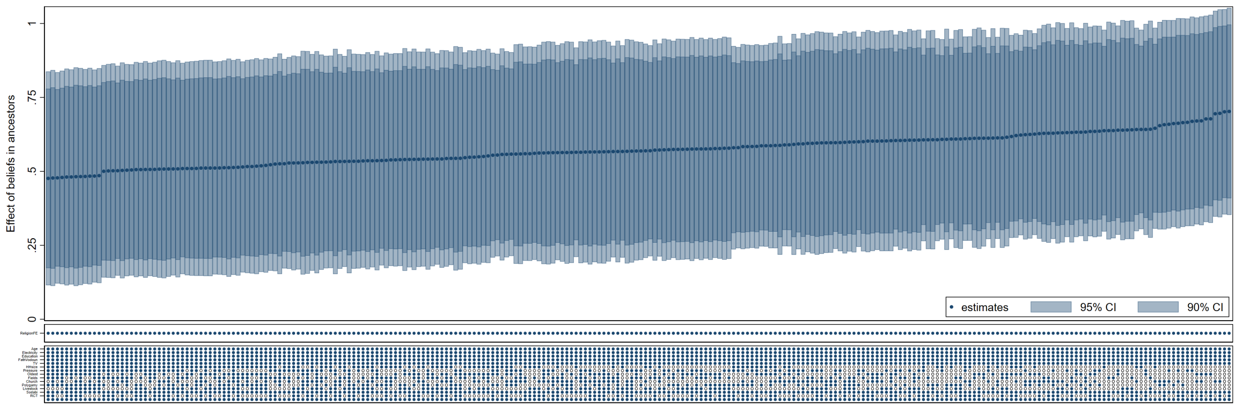
Money to church to get protection against COVID: I try to account for strong religiosity (beyond the inclusion of religion fixed effects) by looking at whether the respondent has donated some money to his church to get protected against COVID (average is 14%).

Drinking sodabi to get protection against COVID: I proxy for the strength of other traditional practices (not related to beliefs in supernatural powers) by measuring the subjective effectiveness of traditional medicine. I include an indicator that equals one if the respondent drunk a traditional alcoholic drink (sodabi) to get protection against COVID.

Livestock ownership: As an additional measure of wealth, I include a dummy variable that takes the value one if the respondent owns livestock.

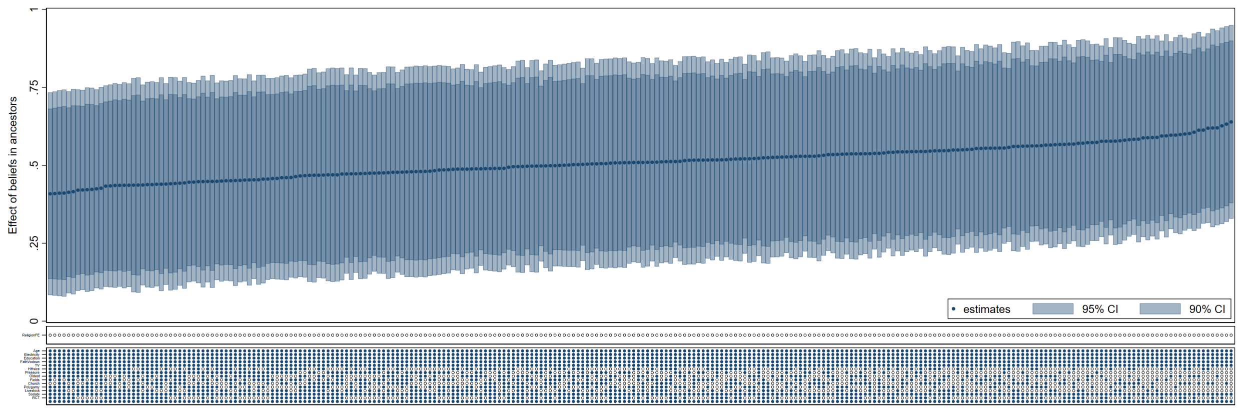
Husband is the oldest brother: Finally, since siblings may compete for limited resources in their household and first-born children are often prioritized in accessing resources, or in their marriage prospects, I include an indicator that takes value one if the respondent's husband is the oldest sibling (among those from the same mother).

Figure G1: Specification curve: additional controls and religion fixed effects



Note: The figure shows the specification curves for the effect of beliefs in the influence of ancestors. Each dot is a coefficient from Eq. 1 with a different set of control variables. The vertical bars, from darkest to lightest, denote the 95% and 90% confidence intervals. All specifications include religion fixed effects. *HHsize* is the total number of people living in the household. *FatherVodoun* equals one if the (main) religion of the husband's father is Voodoo and zero otherwise. *Oldest* takes the value one if the respondent is the oldest brother and zero otherwise. *Church* equals one if the respondent donated to his church to get protected from COVID. *Polygamy* equals one if the respondent is in a polygamous union and zero otherwise. *Livestock* is an indicator that takes the value one if the respondent owns livestock. *Sodabi* takes the value one if the respondent drank sodabi (a traditional alcoholic drink) to get protected against COVID.

Figure G2: Specification curve: additional controls

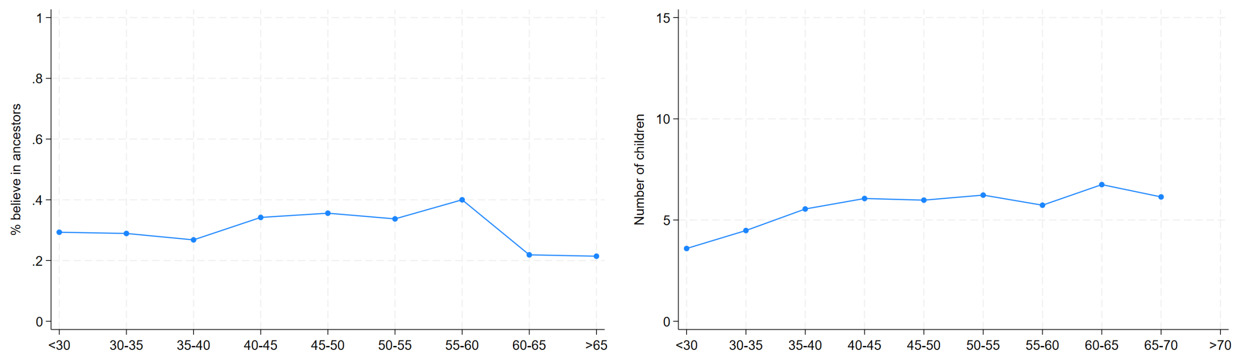


Note: The figure shows the specification curves for the effect of beliefs in the influence of ancestors. Each dot is a coefficient from Eq. 1 with a different set of control variables. The vertical bars, from darkest to lightest, denote the 95% and 90% confidence intervals. Religion fixed effects are not included. *HHsize* is the total number of people living in the household. *FatherVodoun* equals one if the (main) religion of the husband's father is Voodoo and zero otherwise. *Oldest* takes the value one if the respondent is the oldest brother and zero otherwise. *Church* equals one if the respondent donated to his church to get protected from COVID. *Polygamy* equals one if the respondent is in a polygamous union and zero otherwise. *Livestock* is an indicator that takes the value one if the respondent owns livestock. *Sodabi* takes the value one if the respondent drank sodabi (a traditional alcoholic drink) to get protected against COVID.

G.6. *Age dynamics: age fixed effects*

One potential confounder could be related to similar dynamics in the evolution of beliefs and the number of children over the life cycle or over time, that are difficult to capture in a lineal model. However, this does not seem to be the case (see Figure G3). Although the number of children is an increasing function of age (although at decreasing pace), the share of women believing in ancestors is rather constant. In Table G8 I show that adding age fixed effects does not change the results.

Figure G3: Dynamics of beliefs in ancestors and children by age group



(a) Beliefs in ancestors

(b) Number of children

Notes: Panel (a) plots the share of people believing in the influence of ancestors on their lives (n=943) by age group. Panel (b) plots the average number of children by age group.

Table G8: Beliefs in ancestors and fertility in Benin, age fixed effects

	Number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	1.848***	1.743***	1.249***	1.460***	1.369***	1.251***
	(0.379)	(0.358)	(0.311)	(0.388)	(0.365)	(0.371)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	No	No	No	Yes	Yes	Yes
Age FE	No	Yes	Yes	No	Yes	Yes
Mean Y	7.995	8.016	8.016	8.012	8.029	8.029
N	846	832	832	841	830	830

NOTE. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Baseline controls include whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple, and whether the respondent has received pressure from the extended family to have children. Age fixed effects included. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

G.7. Partible vs Impartible inheritance

Differences in inheritance rules may have an influence on fertility behavior. For example, impartible inheritance (where a single heir receives the full amount of the family's land) has been shown to increase fertility rates (when comparing with partible inheritance rules) both in Europe (Gay et al., 2023) and in sub-Saharan Africa (Fontenay et al., 2024). In Benin, there are ethnic groups with both partible (Yoruba) and impartible inheritance (Fon and Adja), so it is an interesting setting to test whether the positive relationship between ancestral beliefs and fertility is affected by differences in inheritance rules. In fact, impartible inheritance may be correlated with ancestral beliefs. For example, Chu (1991) has shown that impartible inheritance (e.g., primogeniture) may emerge in traditional societies as the family's head optimal policy when the objective function is the continuation of the family line, which as I have shown is particularly important in societies with strong ancestral beliefs. Therefore, impartible inheritance could be a mediator in the relationship between ancestral beliefs and fertility behavior.

I conduct two different tests to rule out this possibility. First, as impartible inheritance is defined at the ethnic group level, I show in Table G9 that my results remain unchanged when I include ethnicity fixed effects. Second, I test the correlation between ancestral beliefs, impartible inheritance and fertility (Table G10). I show that there is positive correlation between ancestral beliefs and impartible inheritance (column 1), pointing in the direction of Chu (1991)'s hypothesis, namely that impartible inheritance may emerge in places where the motive to continue the family line plays an central role. However, there is not significant relationship between impartible inheritance and fertility (see column 2). Finally, when I include both ancestral beliefs and impartible inheritance, only ancestral beliefs matter for fertility behavior. Overall, these results suggest that, although impartible inheritance and the motive to continue the family line may be related, the positive influence on fertility is mostly due to the later in this context.

Table G9: Beliefs in ancestors, female fertility, and ethnicity fixed effects

	Total number of children ever born					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.825***	0.696***	0.701***	0.896***	0.719***	0.728***
	(0.198)	(0.180)	(0.177)	(0.202)	(0.184)	(0.181)
Baseline controls	No	Yes	Yes	No	Yes	Yes
Extended controls	No	No	Yes	No	No	Yes
Religion FE	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	5.365	5.352	5.352	5.366	5.353	5.353
R-Squared Y	0.0534	0.272	0.292	0.0586	0.275	0.295
N	846	840	840	844	838	838

NOTE. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the respondent answers *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?* Baseline controls include age, age squared, whether the household has electricity and education. Extended controls include the number of agricultural fields, whether the household owns a TV, whether the household has been part of a program to encourage the production of pineapple. Religion fixed effects includes five categories: Voodoo, Roman Catholic, Evangelic, Celeste and other. Ethnicity fixed effects include Adja, Fon, Yoruba and Other. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Table G10: Beliefs in ancestors, impartible inheritance, and fertility

	Total number of children ever born			
	(1)	(2)	(3)	(4)
Ancestral beliefs	0.0664***		0.878***	0.722***
	(0.0174)		(0.214)	(0.193)
Impartible inheritance		0.225	-0.0388	0.203
		(0.241)	(0.276)	(0.234)
Controls	No	No	No	Yes
Religion FE	Yes	Yes	Yes	Yes
Mean Y	0.926	5.302	5.323	5.312
R-Squared Y	0.152	0.0316	0.0565	0.302
N	767	860	767	763

Note. Data: First-hand data collected in southern Benin. The table reports OLS estimates. The outcome variable is whether the ethnic group practiced impartible inheritance in column (1), and the total number of children ever born in columns (2), (3), (4) and (5). "Ancestral beliefs" is a dummy variable that equals one if the respondent believes that the spirits of his ancestors influence his life. "Impartible inheritance" equals one if the respondent's ethnic group practiced impartible inheritance. Controls include age, whether the household has electricity, education, number of agricultural fields, whether the household owns a TV, and whether the household has been part of a program to encourage the production of pineapple. Robust standard errors in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

G.8. Old-age security motive for fertility

Ancestor worship could in part be interpreted as an extension of the well-known "old-age security motive", whereby people's needs for old-age support raise the demand for children (Lambert and Rossi, 2016; Rossi and Godard, 2022; Sage, 2023). In fact, in societies where the continuation of the family line is of central importance due to the influence of ancestors, the question of "security" has two different facets (Goody, 1973). In addition to the standard security in old age, there is the security in the after-life which drives the demand for children in the same way as the former. Ancestors would need their descendants to sustain themselves in the afterlife as much as the living need them in old age. It is therefore important to distinguish between security in old age and security in the after-life. I use detailed field-level information on property rights to show that my results are not affected when I control for the quantity of owned (and rented) land by women (both in terms of area in squared meters and of number).

Land ownership should be negatively correlated with fertility when the old-age security motive for fertility is important (in my sample, the median number of owned fields by women is one, and the median area of owned fields is 800 squared meters).

Table G11: Spouses' beliefs in ancestors, land ownership and female fertility

	Number of children ever born			
	(1)	(2)	(3)	(4)
Ancestral beliefs	0.698***	0.694***	0.714***	0.693***
	(0.179)	(0.178)	(0.179)	(0.178)
Controls	Yes	Yes	Yes	Yes
Area fields	No	Yes	No	Yes
Number fields	No	No	Yes	No
Religion FE	Yes	Yes	Yes	Yes
Mean Y	5.352	5.352	5.352	5.352
N	840	840	840	840

Note. Data: First-hand data collected in southern Benin. The sample is restricted to women. The table reports OLS estimates. The outcome variable is the total number of children ever born. "Men + Women beliefs" is a dummy variable that equals one if both the husband and the wife answer *yes* to the question *Do you believe that the spirits of your ancestors influence the events of your life?*. Column 1 does not include any control related to land. Control 2 controls for the total area of all owned and rented fields. Column 3 controls for the total number of owned and rented fields. Column 4 control for the total area of all fields. Controls include age, age squared, whether the household has electricity, education, whether the household owns a TV, and whether the respondent has received pressure from the extended family to have children. Religion fixed effects includes five categories: Vodoun, Roman Catholic, Evangelic, Celeste and other. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

G.9. Robustness Tests for Selection on Unobservables

G.9.1. Altonji, Elder, and Taber (2005) Test

The Altonji et al. (2005)'s test evaluates the robustness of treatment effects to potential omitted variable bias by comparing the degree of selection on observables to the hypothetical selection on unobservables required to eliminate the estimated effect. The test is based on the intuition that the amount of bias from observed covariates provides information about the likely magnitude of bias from unobserved factors. The Altonji ratio is calculated as:

$$\text{Ratio} = \beta_{\text{controlled}} / (\beta_{\text{uncontrolled}} - \beta_{\text{controlled})}$$

where $\beta_{\text{uncontrolled}}$ is the treatment coefficient from a regression without controls and $\beta_{\text{controlled})}$ is the coefficient from the full specification with all covariates. The ratio indicates how many times stronger selection on unobservables would need to be, relative to selection on observables, to drive the treatment effect to zero. Values substantially greater than one suggest that implausibly strong unobserved confounding would be required to eliminate the estimated effect. The key assumption underlying this test is that selection on observables provides a reasonable benchmark for the likely strength of selection on unobservables.

G.9.2. Oster (2019) Extension

Oster (2019) Oster (2019) extends the Altonji framework by incorporating information about R-squared movements and allowing researchers to make explicit assumptions about the maximum achievable R-squared (Rmax) in a hypothetical regression with all relevant variables observed. Two assumptions are required. First, a value for the relative degree of selection on observed and unobserved variables is needed (δ). Second, a value for the R-squared from a hypothetical regression of the outcome on treatment and both observed and unobserved controls (R_{max}).

Regarding δ, Oster (2019) suggests equal selection ($\delta = 1$) as an appropriate upper bound on δ. A value of $\delta = 1$ suggests that observables are at least as important as the unobservables. This assumptions seems likely to hold since observed variables are (usually) the factors that are most important. Regarding R_{max}, different bounds that are a function of the R-squared from the regression with observed controls (R') can be chosen. Based on a sample of randomized articles, Oster (2019) suggests $R_{\text{max}} = 1.3 \times R'$, which would allow about 90% of randomized results to survive (but only 45% of nonrandomized results).[68]

The Oster bounds are calculated as:

$$\tilde{\beta} = \beta_{\text{controlled}} - \delta \times (\beta_{\text{uncontrolled}} - \beta_{\text{controlled}}) \times \left[\frac{R^2_{\text{max}} - R^2_{\text{controlled}}}{R^2_{\text{controlled}} - R^2_{\text{uncontrolled}}} \right]$$

where δ represents the relative degree of selection on unobservables compared to observables, and R_{max} is the maximum R-squared achievable with perfect controls. The parameter δ^* indicates the critical value of δ that would drive the treatment effect to zero under proportional selection assumptions. Oster's approach improves upon the original Altonji test by explicitly modeling the relationship between coefficient movements and explanatory power, providing bounds on treatment effects under different

[68]There are good reasons to use thresholds below 1. For example, there might be measurement error in the dependent variable, as it is probably the case here.

assumptions about the strength of unobserved confounding. Values of δ^* substantially greater than one indicate that unobservables would need to be much more important than observables to explain away the results, suggesting robustness to omitted variable bias.

H. Democratic Republic of Congo: Robustness

H.1. Regression model

The equation I estimate takes the following form (with small variations depending on the specification). Let i denote individuals, e denote ethnicity, and c denote city of residence.

$$Y_{iect} = \beta_0 + \beta_1 AncestralBeliefs_e + X_i'\Phi + \alpha_c + \epsilon_{iect} \tag{6}$$

Where Y_{iect} is a fertility outcome, $AncestralBeliefs_e$ is a dummy variable that equals one if the respondent belongs to an ethnic group that traditionally practiced ancestor worship. The vector X_i' includes an extensive set of individual-level control variables, such as age, age squared, whether the place of birth was rural or urban, education, whether the respondent is a migrant, whether respondent's father is alive at the time of the survey, employment status, whether the respondent works in agriculture (only available for the urban sample), and year of installation in the current city. Finally, α_c denotes city of residence fixed effects. ϵ is an error term, and standard errors are clustered at the ethnic group level, since the treatment is constructed at that level.

The presence of city of residency fixed effects means that, when estimating equation 6, the influence of ancestor worship is identified by comparing respondents living in the sample place (and therefore facing the same supply constraints) but who migrated from different areas and therefore have different belief systems.[69]

[69] This strategy mirrors the epidemiological approach introduced by Fernández and Fogli (2009), who compare the fertility outcomes of second-generation American women (to hold market and institutions constant) to look at the effect of culture (proxied by their parents' country of origin). The empirical strategy here differs in several aspects. First, an important limitation is that I do not have information on second-generation migrants, and therefore rely on first-generation migrants. However, an important advantage is that I can control for the year of installation in the city of residence at the time of the survey, which allows me to eliminate concerns related to a potential correlation between time since migration and the prevalence of ancestral beliefs, which could confound identification and is often ignored in the literature (Bertoli et al., 2024).

H.2. *Age-specific fertility rates*

I show that my results hold when I use age-specific fertility rates instead of total number of children ever born. Using birth calendars, I reconstruct the total number of births women had at ages 35, 40, and 45. The results are reported in Table H1. While fertility is more likely to be completed at older ages, the sample size becomes smaller because there are fewer women in the older age categories. I find a positive effect of ancestor worship on age-specific fertility, ranging from an average effect of 18% of the full sample mean at age 35 to 21% at age 40 and 26% at age 45. Here, I observe an stronger effect on the rural sample (again, note that I include city of residence FE when using the urban sample and region FE when I use the EDOZA sample, which can explain – at least partially – the stronger effect on the rural sample. I find that ancestor worship increases the number of births at age 30 by 27% in the EDOZA sample and by 15% in the urban sample. At age 40, the effect moves up to 31% of the EDOZA sample mean and to 18% in the urban sample. Finally, at age 45, ancestor worship is associated with an increase of 36% of the sample mean in the EDOZA survey and of 23% in the urban sample.

Table H1: Ancestral beliefs and age-specific fertility in the DRC

	Births at 35			Births at 40			Births at 45		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Ancestral beliefs	0.767***	1.139***	0.692***	0.984***	1.358***	0.902***	1.193***	1.583***	1.113***
	(0.172)	(0.349)	(0.165)	(0.229)	(0.403)	(0.250)	(0.244)	(0.402)	(0.259)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
All	Yes	No	No	Yes	No	No	Yes	No	No
EDOZA	No	Yes	No	No	Yes	No	No	Yes	No
Urban	No	No	Yes	No	No	Yes	No	No	Yes
Mean Y	4.568	4.243	4.746	4.862	4.580	5.055	4.710	4.489	4.890
R-Squared	0.246	0.256	0.274	0.244	0.267	0.267	0.213	0.252	0.226
N	18997	6730	12267	13447	5466	7981	9678	4337	5341

NOTE. Data: Demographic Survey of 1970s and EDOZA. The table reports OLS estimates. The outcome variables are the total number of births a women had before age 35 (columns 1, 2 and 3), 40 (columns 4, 5 and 6) and 45 (columns 7, 8 and 9). "Ancestor worship" is a dummy variable that equals one if the ethnic group e practice ancestor worship. Controls include age, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

H.3. Intensive vs. Extensive margin and estimation of Poisson model

This subsection shows that the results are not affected when looking at both the extensive and the intensive margin, and are robust to the use of a pseudo-poisson maximum-likelihood estimator (see subsection H.3). This is important, as the dependent variable contains a large proportion of zeros (37% of women over the age of 13 in my sample). This is a well know fact in certain areas of Central Africa, known as the "infertility belt", specially before the 1990s (Bongaarts et al., 1984; Larsen, 2003), where infertility rates among married women could reach 25%.

Table H2: Ancestral beliefs and fertility in the DRC, Poisson regression

	Total number of children ever born					
	Full sample		EDOZA		Urban sample	
	+30	+40	+30	+40	+30	+40
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.188***	0.249***	0.218***	0.284***	0.186***	0.276***
	(0.0434)	(0.0533)	(0.0731)	(0.0651)	(0.0358)	(0.0622)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	5.322	5.080	4.865	4.693	5.542	5.359
N	23194	12435	7551	5207	15643	7228

NOTE. Data: Urban Demographic Survey of 1970s and EDOZA (rural and urban areas). Columns 1 and 2 combine both the Urban Demographic survey and EDOZA. Columns 3 and 4 use only the EDOZA sample. Columns 5 and 6 use only the urban sample. In columns 2, 4, 6, the sample is restricted to women older than 30 years old. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the total number of children ever born. "Ancestral beliefs" is a dummy variable that equals one if the ethnic group e practiced ancestor worship before European colonization. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City fixed effects are included for the urban sample and region fixed effects included in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

Table H3: Ancestral beliefs and fertility in the DRC, extensive vs intensive margin

	Total number of children ever born					
	Full sample		EDOZA		Urban sample	
	Has children	Number of children	Has children	Number of children	Has children	Number of children
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.0701***	0.585***	0.0820***	0.771**	0.0766***	0.509***
	(0.0127)	(0.182)	(0.0173)	(0.345)	(0.0146)	(0.133)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes		Yes	Yes	Yes
City fixed effects	Yes	Yes		Yes	Yes	Yes
Extensive margin	Yes	No	Yes	No	Yes	No
Mean Y	0.882	6.036	0.840	5.793	0.902	6.146
R-Squared	0.112	0.214	0.151	0.196	0.0937	0.240
N	23194	20448	7551	6342	15643	14106

NOTE. Data: Demographic Survey of 1970s and EDOZA. The sample is restricted to women older than 30 years old. Columns 1 and 2 combine both the Urban Demographic survey and EDOZA. Columns 3 and 4 use only the EDOZA sample. Columns 5-6 use only the urban sample. Columns 1, 3, and 5 look at the extensive margin (probability of having children), while columns 2, 4, and 6 look at the intensive margin (total number of children ever born conditioned on having children). The table reports OLS estimates. "Ancestor worship" is a dummy variable that equals one if the ethnic group *e* practice ancestor worship. Controls include age, age squared, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

H.4. Unobserved human capital

Here, I show that the results are not driven by unobserved differences in human capital. For example, the results could be biased if societies in which ancestor worship is important may have been less receptive to missionaries, which could have lowered the human capital of the ethnic network, broadly defined, relative to neighboring groups, affecting the probability of marriage or obtaining a job. To address this concern, I examine whether ancestor worship is related to wages. Indeed, if unobserved human capital accounts for part of the results, it should be reflected in wages. To do this, I resort to an expenditure survey which includes 1/50 of the total number of households identified in the same seven cities of the 1970s Urban Demographic Survey.[70] Table H4 reports the results from the regression of wages on ancestor worship and a vector of control variables, including education. Ancestor worship remains insignificant in every specification, suggesting that unobserved human capital is not responsible for the results.

[70]See Appendix F for details about this survey.

Table H4: Ancestral beliefs and unobserved human capital in the DRC

	Pseudo-Poisson Maximul Likelihood	OLS estimates	
	Total wage	Log(1+ total wage)	Total wage
	(1)	(2)	(3)
Ancestor Worship	-0.107	0.236	-949.7
	(0.208)	(0.305)	(1315.8)
Years of education	0.161***	0.297***	628.8***
	(0.0144)	(0.0353)	(92.69)
City FE	Yes	Yes	Yes
Territory of birth FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Mean Y	4047.8	5.411	4045.2
R-squared		0.234	0.191
N	15708	15718	15718

Note. Data: Budgetary Survey of the 1970s and Vansina(1966). Column 1 reports Pseudo-Poisson Maximul Likelihood estimates, and the dependent variable is the total wage. Column 2 reports OLS estimates, and the dependent variable has been transformed to Log(1+wage). Finally, column 3 reports OLS estimates using total wage as dependent variable. Ancestor worship is a dummy variable that equals one if the ethnic group e of individual i traditionally practices ancestor worship. Controls include age, age squared, total number of household members, gender, whether respondents were born in urban/rural area, education, dummy equal to one if migrant, year of installation in the current city, dummy equal to one if the father's respondent is alive, and dummy equal to one if the respondent is a farmer. Standard errors () are clustered at the ethnic group level. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

H.5. Exposure to missionary presence

An important limitation of the 1970s datasets is that they do not contain any information on religious beliefs. More importantly, the rapid expansion of Christianity in the context of the DRC took place hand in hand with missionaries' activities, which may have weakened family and ethnic lineages, and emphasized universal over family/ethnicity-centered moral values (Platteau, 2009; Reybrouck, 2014; Bergeron, 2025).[71] At the same time, exposure to missionary presence in colonial Congo had important effects on education (Alvarez-Aragon et al., 2025) and fertility outcomes (Guirkinger and Villar, 2022).

I construct a measure of exposure to all Christian missions opened in the DRC between 1885 and 1948 for all individuals born before 1948 using the West Zaire Demographic surveys and data on missions from Guirkinger and Villar (2022) and Alvarez-Aragon

[71]The relationship between missionaries and the practice of ancestor worship is controversial. While some authors suggest that Christian and Islamic doctrines prohibited ancestor worship, other authors suggest that ancestor worship is often allowed to continue because it is considered a cultural rather than a religious practice (Caldwell and Caldwell, 1990; Olupona, 2014). In fact, this is what I observe in Benin: there is no correlation between being Catholic and beliefs in the influence of ancestors.

et al. (2025). In the historical individual-level data, information on the territory of birth of each individual is available, so we can construct the exposure to missionary presence of each territory in each year and assign this measure to each respondent according to their year of birth. To do this, I follow Guirkinger and Villar (2022) and Alvarez-Aragon et al. (2025) and construct a continuous measure of proximity at the territory level that controls for mission density. Specifically, I generate 1000 random points within each territory and compute the distance from each random point to its closest mission before averaging over these distances. This process is repeated for each territory, each year between 1885 and 1948, and three types of missions: Catholic, Catholic with nuns, and Protestant. When showing my results, I report negative (log) distances to have an easier interpretation.

As we can observe in Table H5, the coefficients barely move when I control for exposure to missionary presence. The magnitude of the effects remain high. Ancestor worship is associated with an increase of 17% in the number of births at age 35 (using the full sample), of 20% age age 40, and of 25% at age 45. Moreover, I am able to replicate the main findings in Guirkinger and Villar (2022). Exposure at birth to Catholic missions with nuns is associated with higher fertility rates, while exposure to Protestant missionaries generally reduces fertility.

Table H5: Ancestral beliefs, missions, and age-specific fertility in the DRC

	Births at 35			Births at 40			BIrths at 45		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Ancestral beliefs	0.736***	1.011***	0.641***	0.943***	1.209***	0.839***	1.133***	1.432***	0.984***
	(0.170)	(0.359)	(0.160)	(0.221)	(0.413)	(0.233)	(0.242)	(0.399)	(0.233)
Exp to Catholics	-0.112	-0.153	0.0199	-0.0308	-0.0971	0.0520	0.0574	-0.151	0.282**
	(0.191)	(0.344)	(0.120)	(0.243)	(0.445)	(0.142)	(0.208)	(0.393)	(0.130)
Exp to Cath nuns	0.234**	0.484**	0.281**	0.254**	0.548***	0.362**	0.271*	0.480**	0.407***
	(0.106)	(0.186)	(0.134)	(0.125)	(0.191)	(0.155)	(0.140)	(0.208)	(0.140)
Exp to Protestants	0.0154	0.230	-0.121**	-0.0489	0.169	-0.202*	-0.0799	0.247	-0.365***
	(0.127)	(0.288)	(0.0587)	(0.164)	(0.371)	(0.103)	(0.212)	(0.433)	(0.115)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City x Territory FE	No	No	No	Yes	No	No	No	Yes	Yes
All	Yes	No	No	Yes	No	No	Yes	No	No
Edoza	No	Yes	No	No	Yes	No	No	Yes	No
Urban	No	No	Yes	No	No	Yes	No	No	Yes
Mean Y	4.575	4.244	4.758	4.871	4.581	5.071	4.723	4.488	4.915
R-squared	0.249	0.262	0.273	0.247	0.274	0.267	0.218	0.259	0.229
N	18852	6698	12154	13328	5435	7893	9579	4306	5273

Note. Data: Demographic Survey of 1970s and EDOZA. The table reports OLS estimates. The outcome variables are the total number of births a women had before age 35 (columns 1, 2 and 3), 40 (columns 4, 5 and 6) and 45 (columns 7, 8 and 9). "Ancestor worship" is a dummy variable that equals one if the ethnic group e practice ancestor worship. Details on how exposure to missions is constructed can be found in Section H.5. Controls include age, whether the place of birth was urban or rural, whether the respondent has primary education, dummy equal to one if migrant, dummy equal to one if the father's respondent is alive, dummy equal to one if the respondent is working, dummy equal to one if the respondent is a farmer (only available in the urban sample), number of household members, and year of installation in the current city. City/region fixed effects include city of residence in the case of the urban sample and region of residence in the case of EDOZA. Standard errors clustered at the Vansina's ethnic group-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I. External validity: ancestral beliefs and fertility in different time periods and contexts

I.1. *Ancestral beliefs and contemporary fertility in 19 Sub-Saharan African countries*

I.1.1. *Regression model*

I estimate a similar model than in equation 1, but including country fixed effects and a different set of control variables that varies by specification.[72] I follow Gershman (2016) and measure ancestral beliefs at both the individual and the regional level.[73] I estimate the following equation, where i denote individuals, r denote regions, and c denote countries:

$$Y_{ic} = \alpha + \beta AncestralBeliefs_{i(r)} + X_i'\Phi + \phi_c + \epsilon_{ic} \quad (7)$$

Y_{ic} is the total number of children ever born of individual i living in country c, $AncestralBeliefs_{i(r)}$ is the main explanatory variable and is measured at two different levels depending on the specification. $AncestralBeliefs_i$ captures personal belief in ancestors (measured as a dummy variable equal to one if the respondent answers "*Yes*" to the question "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*"), while $AncestralBeliefs_r$ measures the prevalence of ancestral beliefs in each respondents region of residence (measured as the fraction of survey participants in a given region who answer "yes" to the aforementioned question). The vector X_i' contains an extensive set of individual-level covariates that vary according to the specification.[74] The analysis exploits within-country variation by including ϕ_c, which represents country fixed effects, to account for any country-level time-invariant

[72] These controls include age, age squared, gender, urban or rural place of residence, religion, education, marital status, and a dummy variable that equals one if the respondent finds himself/herself in a good economic situation.

[73] Ancestral Beliefs are measured, at the individual level, as a dummy variable equal to one if the respondent answers "*Yes*" to the question "*Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?*". At the regional level, they are measured as the prevalence of ancestral beliefs in each respondents region of residence (measured as the fraction of survey participants in a given region who answer "yes" to the aforementioned question).

[74] These include age, age squared, gender, whether the respondent lives in a urban or rural area, whether the respondent is Christian/Muslim or from another religion, a dummy variable that equals one if the respondent has completed primary education, a dummy variable that equals one if the respondent is married, and a dummy variable that equals one if the respondent finds himself/herself in a good economic situation.

characteristics that may be correlated with both fertility levels and traditional beliefs. Robust standard errors are included in parenthesis.

I.1.2. Results

The results are shown in Table I1. I find a positive and robust relationship between ancestral beliefs and fertility outcomes. In terms of magnitude, the coefficient ranges from 0.5 (25% of the outcome mean) more children in column (1) to about 0.2 more children (8% of the outcome mean) in column (4), once we include the full set of control variables. This effect is similar to the coefficient associated with living in a rural area (0.2) and almost the same as the influence of having less than primary education (0.3). Interestingly, the prevalence of ancestral beliefs at the regional level strongly correlates positively with fertility rates at the individual level.

Table I1: Ancestral beliefs and fertility in sub-Saharan Africa

	Number of children ever born							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral Beliefs	0.507***	0.248***	0.233***	0.172***				
	(0.0378)	(0.0302)	(0.0301)	(0.0292)				
Ancestral beliefs (region)					0.472***	0.611***	0.550***	0.335***
					(0.136)	(0.103)	(0.103)	(0.0995)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended Controls 1	No	No	Yes	Yes	No	No	Yes	Yes
Extended Controls 2	No	No	No	Yes	No	No	No	Yes
Mean Y	2.092	2.086	2.086	2.096	2.093	2.087	2.087	2.097
R-squared	0.0635	0.417	0.424	0.480	0.0559	0.415	0.422	0.478
N	22926	22791	22791	21997	23394	23258	23258	22386

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variable is the total number of children ever born. "Ancestral beliefs" is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". "Ancestral beliefs (region)" is the regional prevalence of ancestral beliefs and it is set equal to the fraction of survey participants in a given region who personally believe in ancestors. Basic controls include age, age^2 and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Robustness.- In Appendix I.1, I show that the results are not driven by potential confounders such as overall religiosity or the importance of alternative traditional magico-religious beliefs (e.g., traditional religious healers, belief in miracles, participation in initiation rituals, beliefs in witchcraft, or knowledge about ancestral, tribal, animist or other traditional African religions), suggesting that the effect is specific to beliefs in ancestors. Moreover, I show that these results are robust to alternative definitions of the explanatory variable, to the inclusion of region (instead of country) fixed effects, or to alterative clustering of the standard errors.[75]

I.1.3. Alternative definitions of "ancestral beliefs"

Table I2: Sacrifices to ancestors and fertility in sub-Saharan Africa

	Number of children ever born							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Sacrifices	0.332***	0.167***	0.149***	0.102***				
	(0.0370)	(0.0294)	(0.0294)	(0.0286)				
Sacrifices to ancestors (region)					0.636***	0.476***	0.493***	0.342***
					(0.161)	(0.123)	(0.122)	(0.117)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended Controls 1	No	No	Yes	Yes	No	No	Yes	Yes
Extended Controls 2	No	No	No	Yes	No	No	No	Yes
Mean Y	2.101	2.095	2.095	2.103	2.093	2.087	2.087	2.097
R-squared	0.0606	0.418	0.425	0.481	0.0560	0.415	0.421	0.478
N	22243	22115	22115	21361	23394	23258	23258	22386

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variable is the total number of children ever born. "Sacrifices to ancestors" is an indicator equal to one if respondent answers "yes" to the following question: "do you believe that sacrifices to spirits or ancestors can protect you from bad things happening?". "Sacrifices to ancestors (region)" is the regional prevalence of ancestral beliefs and it is set equal to the fraction of survey participants in a given region who personally believe in the influence to making sacrifices to ancestors. Basic controls include age, age^2 and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

[75] Since the sample design in Pew Research Center (2009)'s survey was a stratified random sample at the regional level (province, region or district depending on the country), an alternative is to cluster the standard errors at the regional level. Table I5 in Appendix I.1 shows that this choice does not matter for my results, even if clustered standard errors at this level may be inflated, since the number of clusters in the sample equals the number of clusters in the population (Abadie et al., 2022).

Table I3: Rituals or sacrifices to ancestors and fertility in sub-Saharan Africa

	Number of children ever born							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Rituals or Sacrifices	0.393***	0.192***	0.180***	0.130***				
	(0.0350)	(0.0279)	(0.0278)	(0.0270)				
Rituals or Sacrifices (region)					0.703***	0.678***	0.681***	0.475***
					(0.139)	(0.107)	(0.107)	(0.103)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended Controls 1	No	No	Yes	Yes	No	No	Yes	Yes
Extended Controls 2	No	No	No	Yes	No	No	No	Yes
Mean Y	2.098	2.093	2.093	2.101	2.093	2.087	2.087	2.097
R-squared	0.0625	0.418	0.425	0.481	0.0564	0.415	0.422	0.478
N	21910	21783	21783	21077	23394	23258	23258	22386

Note. Data: PEW research forum 2008-2009 Survey. The outcome variable is the total number of children ever born. "Rituals or Sacrifices" is an indicator equal to one if respondent answers "yes" to at least one of the following questions: "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?" or "Do you believe that sacrifices to spirits or ancestors can protect you from bad things happening?". "Rituals or Sacrifices (region)" is the regional prevalence of ancestral beliefs and it is set equal to the fraction of survey participants in a given region who personally conduct rituals to honour ancestors or believe in the influence of making sacrifices to ancestors. Basic controls include age, age^2 and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I.1.4. Additional controls: region fixed effects and clustered standard errors

Table I4: Ancestral beliefs and fertility in sub-Saharan Africa, region fixed effects

	Number of children				P(≥ 5 children)			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.520***	0.211***	0.200***	0.143***	0.0369***	0.0119**	0.0103**	0.00713
	(0.0407)	(0.0324)	(0.0324)	(0.0315)	(0.00534)	(0.00493)	(0.00489)	(0.00497)
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Basic Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended Controls 1	No	No	Yes	Yes	No	No	Yes	Yes
Extended Controls 2	No	No	No	Yes	No	No	No	Yes
Mean Y	2.100	2.095	2.095	2.104	0.0997	0.0993	0.0993	0.0993
R-squared	0.0982	0.442	0.445	0.501	0.0684	0.201	0.204	0.212
N	21980	21845	21845	21068	21980	21845	21845	21068

Note. Data: PEW research forum 2008-2009 Survey. The outcome variables are defined as follows: in columns (1)-(4), it is the total number of children, and in columns (5)-(8) it is the probability of having at least 5 children. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Basic controls include age, age^2 and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I5: Ancestral beliefs and fertility in sub-Saharan Africa, alternative clustering

	Number of children ever born							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral Beliefs	0.507***	0.248***	0.233***	0.172***				
	(0.0568)	(0.0382)	(0.0375)	(0.0374)				
Ancestral beliefs (region)					0.472*	0.611***	0.550***	0.335*
					(0.281)	(0.218)	(0.206)	(0.184)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Extended Controls 1	No	No	Yes	Yes	No	No	Yes	Yes
Extended Controls 2	No	No	No	Yes	No	No	No	Yes
Mean Y	2.092	2.086	2.086	2.096	2.093	2.087	2.087	2.097
R-squared	0.0635	0.417	0.424	0.480	0.0559	0.415	0.422	0.478
N	22926	22791	22791	21997	23394	23258	23258	22386

Note. Data: PEW research forum 2008-2009 Survey. The outcome variable is the total number of children ever born. "Ancestral beliefs" is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". "Ancestral beliefs (region)" is the regional prevalence of ancestral beliefs and it is set equal to the fraction of survey participants in a given region who personally believe in ancestors. Basic controls include age, age^2 and sex. Extended controls 1 also include whether the individual lives in a urban or rural area and whether he/she is christian/muslim or from other religion. Finally, extended controls 2 also include a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Standard errors clustered at the region-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I.1.5. Additional controls: religiosity and supernatural beliefs

In this subsection, I show that the results are not driven by overall religiosity or alternative magico-religious beliefs. Indeed, religiosity has been shown to be positively related to fertility (Hayford and Morgan, 2008; Herzer, 2019). Therefore, even though religion is controlled for in all specifications, the results would still be biased if the measure of ancestral beliefs is correlated with residual religiosity (not captured by religious affiliation). Table I6 shows that the coefficient on ancestor worship remains unchanged after controlling for the frequency of religious service attendance (Herzer, 2019) and the importance of religion in respondents' lives (Hayford and Morgan, 2008), which are the two most common ways to measure religiosity in the literature. A second potential confounder relates to the importance of alternative traditional magical-religious beliefs, which have been shown to have an important influence on economic life (Platteau, 2014; Le Rossignol et al., 2022). Table I7 shows that alternative measures of traditional supernatural beliefs, such as the use of traditional religious healers, belief in miracles, participation in initiation rituals, beliefs in witchcraft, or knowledge about ancestral, tribal, animist or other traditional African religions, do not affect the results, suggesting that my proxies effectively capture something related specifically to the belief in ancestors.

Table I6: Ancestral beliefs and fertility, controlling for overall religiosity

	Number of children ever born		
	(1)	(2)	(3)
Ancestral beliefs	0.184***	0.183***	0.184***
	(0.0292)	(0.0291)	(0.0292)
High religious attendance		0.138***	0.136***
		(0.0459)	(0.0461)
Religion is important			0.0254
			(0.0343)
Country FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
Mean Y	2.083	2.083	2.083
R-squared	0.480	0.481	0.481
N	22295	22295	22295

NOTE. Data: PEW research forum 2008-2009 Survey. The outcome variables is the total number of children ever born. The explanatory variable is an indicator equal to one if the respondent answers "a great deal" or "some" to the question "How much would you say you know about ancestral, tribal, animist, or other traditional African religions?". *Attendance is a dummy variable that equals one if the respondent attends more than once a week religious events*. *Importance* is a dummy variable that equals one if the respondent declares than religion is "very important" in his/her life. Controls include age, age^2, gender, whether the individual lives in a urban or rural area, whether he/she is christian/muslim or from other religion, a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I7: Ancestral beliefs and fertility, controlling for other supernatural beliefs

	Number of children ever born									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Ancestral beliefs	0.395***	0.495***	0.370***	0.476***	0.479***	0.143***	0.175***	0.157***	0.163***	0.186***
	(0.0411)	(0.0381)	(0.0431)	(0.0392)	(0.0392)	(0.0318)	(0.0294)	(0.0332)	(0.0301)	(0.0300)
Religious healers	0.261***					0.0766***				
	(0.0356)					(0.0273)				
Experienced miracle		0.131***					0.0580**			
		(0.0315)					(0.0243)			
Initiation ritual			0.282***					0.0380		
			(0.0422)					(0.0322)		
Witchcraft				0.0880**					0.0162	
				(0.0347)					(0.0265)	
Traditional religion					0.221***					0.0447
					(0.0358)					(0.0277)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Mean Y	2.097	2.088	2.091	2.090	2.078	2.099	2.092	2.094	2.093	2.078
R-squared	0.0673	0.0638	0.0660	0.0641	0.0669	0.481	0.482	0.481	0.483	0.486
N	22157	22654	22698	21974	21396	21324	21761	21808	21129	20587

NOTE. Data: PEW research forum 2008-2009 Survey. "Religious healers" equals one if the respondent's family has ever used traditional religious healers when someone is sick. "Experienced miracle" equals one if the respondent has experienced/witnessed a divine healing of an illness or injury. "Initiation ritual" equals one if the respondent has ever participated in an initiation ritual for friends, relatives or neighbors. "Witchcraft" equals one if the respondent declares that he/she believes in witchcraft. "Traditional religion" is a variable that equals one if the respondent answers "*some*" or "*a great deal*" to the question "*How much would you say you know about ancestral, tribal, animist, or other traditional African religions?*. Controls include age, age^2, sex, whether the individual lives in a urban or rural area, whether he/she is christian/muslim or from other religion, a dummy variable that equals one if the respondent has completed primary education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I.2. Ancestral beliefs and fertility in folklore

I.2.1. Regression model

Due to the presence of zeros in the dependent variable and the skewness in its distribution, I follow the seminal paper of Silva and Tenreyro (2006) and use a pseudo-poisson maximum-likelihood estimator, which is well behaved when the proportion of zeros in the dependent variable is large (Gourieroux et al., 1984a,b; Santos Silva and Tenreyro, 2011).[76] I estimate the following model, where e indexes ethnicity:

$$Y_e = \exp\left(\alpha + \beta AncestralBeliefs_e + X_e'\Phi\right)\epsilon_e \tag{8}$$

Where Y_e is the share of folkloric motifs in the oral tradition of an ethnic group related

[76]In their paper analyzing the Folklore data, Michalopoulos and Xue (2021) transform their dependent variables using log(0.01+Y) to account for the skewed nature of concept intensity across oral traditions. However, these "log-like" transformations have well known problems (Chen and Roth, 2023). Table I12shows that the results do not change when I estimate a linear regression model by ordinary least squares.

to the word *birth* in the main specification, and $AncestralBeliefs_e$ is the share of folkloric motifs in the oral tradition of an ethnic group related to the word *ancestors*. The vector X'_e includes ethnicity-level controls divided in three categories. The folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources used to identify those motifs, and the earliest year of publication in the group's oral tradition. The vector of ethnographic controls includes domestic organization around extended families and the presence of segmented communities and localized clans (according to Enke (2019), these are two key characteristics that reflect strong extended family networks, and therefore help me to disentangle the influence of an extended supportive kinship network from the importance of ancestral beliefs), whether patrilineality is the main descent type, whether the group has partible or impartible inheritance, political complexity, the prevalence of monogamy, the importance of pastoralism, the use of the plow, and the proportion of motifs related to the word "supernatural" (to disentangle the influence of ancestor worship from the importance of alternative supernatural beliefs). Finally, I also control for a number of geographic covariates, including tropical climate, precipitation, ruggedness, land quality (population-weighted), and agricultural suitability. Continent fixed effects are always included when I focus on the global sample. Appendix E provides a detailed description of how these variables were constructed.

I.2.2. Results

Table I8 shows the conditional correlations. I find a positive relationship between the share of motifs in the oral tradition of ethnic groups related to ancestors and the share of motifs related to birth at both the global and African levels, suggesting that ancestral beliefs may have contributed to historically high fertility levels worldwide (Caldwell and Caldwell, 1987). One percentage point increase in the share of motifs related to ancestors increase the share of motifs related to birth by about 5%. These results are robust to the inclusion of an extensive set of control variables and to alternative measures of both fertility and ancestral beliefs (see Appendix I.2).

Table I8: Ancestor-related motifs and birth-related motifs in Folklore, Poisson regression

	Share of motifs related to *birth*							
	SSA Sample				Global Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.0517***	0.0484***	0.0389***	0.0376***	0.0499***	0.0472***	0.0472***	0.0460***
	(0.00324)	(0.0109)	(0.0136)	(0.0124)	(0.00360)	(0.00468)	(0.00625)	(0.00655)
SSA Sample	Yes	Yes	Yes	Yes	No	No	No	No
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	4.177	4.177	3.895	3.908	4.086	4.086	3.877	3.848
N	407	407	307	306	1245	1245	951	862

Note. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I.2.3. Ancestral beliefs and fertility in folklore: robustness

Table I9: Ancestor worship related motifs and birth-related motifs in Folklore

	Share of motifs related to *birth*							
	SSA Sample				Global Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.0273***	0.0276***	0.0232***	0.0229***	0.0272***	0.0261***	0.0262***	0.0255***
	(0.00134)	(0.00423)	(0.00528)	(0.00503)	(0.00137)	(0.00191)	(0.00264)	(0.00300)
SSA Sample	Yes	Yes	Yes	Yes	No	No	No	No
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	4.177	4.177	3.895	3.908	4.086	4.086	3.877	3.848
N	407	407	307	306	1245	1245	951	862

Note. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestor worship" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I10: Ancestor-related motifs and fertility-related motifs in Folklore, Poisson regression

	Share of motifs related to *fertility*							
	SSA Sample				Global Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.148***	0.153*	0.203***	0.215***	0.126***	0.112***	0.0992***	0.104***
	(0.00650)	(0.0841)	(0.0658)	(0.0674)	(0.00466)	(0.00621)	(0.00696)	(0.00756)
SSA Sample	Yes	Yes	Yes	Yes	No	No	No	No
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	0.674	0.674	0.562	0.564	0.729	0.729	0.687	0.660
N	407	407	304	303	1245	1245	951	862

Note. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "fertility" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I11: Ancestor worship related motifs and fertility-related motifs in Folklore

	Share of motifs related to *fertility*							
	SSA Sample				Global Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.0745***	0.0746***	0.139***	0.143***	0.0701***	0.0635***	0.0570***	0.0591***
	(0.00305)	(0.0239)	(0.0441)	(0.0479)	(0.00229)	(0.00291)	(0.00369)	(0.00405)
SSA Sample	Yes	Yes	Yes	Yes	No	No	No	No
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	0.674	0.674	0.562	0.564	0.729	0.729	0.687	0.660
N	407	407	304	303	1245	1245	951	862

Note. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports Pseudo-Poisson Maximum Likelihood (PPML) estimators. The outcome variable is the share of motifs related to "fertility" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestor worship" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I12: Ancestor-related motifs and birth-related motifs in Folklore, linear regression

	Share of motifs related to *birth*							
	SSA Sample				Global Sample			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ancestral beliefs	0.619***	0.609***	0.541***	0.536***	0.472***	0.461***	0.420***	0.432***
	(0.128)	(0.133)	(0.156)	(0.156)	(0.116)	(0.117)	(0.141)	(0.147)
SSA Sample	Yes	Yes	Yes	Yes	No	No	No	No
Folklore controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Ethnographic controls	No	No	Yes	Yes	No	No	Yes	Yes
Geographic controls	No	No	No	Yes	No	No	No	Yes
Mean Y	4.177	4.177	3.895	3.908	4.086	4.086	3.877	3.848
N	407	407	307	306	1245	1245	951	862

NOTE. Data: Ethnographic Atlas and Folklore. In columns (1)-(4), the sample is restricted to SSA. The table reports OLS estimates. The outcome variable is the share of motifs related to "birth" in the oral tradition of an ethnic group. "Ancestor worship" is the share of motifs related to "ancestors" in an ethnic group's oral tradition. Folklore controls include the total number of motifs in an ethnic group's oral tradition, the number of publishers of the sources in the group's oral tradition, and the earliest year of publication in the group's oral tradition. Ethnographic controls include whether the domestic organization is around independent nuclear families, whether people are part of localized clans that live as segmented communities, whether the ethnic group is patrilineal, impartible inheritance, political complexity, whether monogamy is dominant, whether the group practices pastoralism, use of historical plough, historical economic development, practice of intensive agriculture, and the share of motifs in an ethnic group related to "supernatural". Geographic controls include tropical climate, precipitations, ruggedness, land quality (population weighted), and agricultural suitability. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

I.3. Contraceptive use

Table I13: Traditional beliefs and contraceptive use

	(1) Currently	(2) Traditional	(3) Modern	(4) Ever	(5) Intention	(6) Fear side-effects	(7) AIDS witchcraft
Traditional beliefs	-0.0268***	-0.00366	-0.0218***	-0.0261***	-0.0342***	-0.0268*	0.0205
	(0.00760)	(0.00378)	(0.00676)	(0.00976)	(0.0120)	(0.0138)	(0.0156)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cluster FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	0.166	0.0592	0.104	0.562	0.382	0.158	0.472
N	43662	43662	43662	43662	36399	14508	27237

NOTE. Data: Demographic and Health Surveys of Benin (1996, 2001, 2006, 2011 and 2017). The table reports OLS estimates. The sample is restricted to women. The outcome variable in column (1) is a dummy that equals one if the respondent was using contraceptives at the time of the survey. In column (2) a dummy that equals one if the respondent was using traditional methods. In column (3) a dummy that equals one if the respondent was using modern methods. In column (4) a dummy that equals one if the respondent has ever used contraceptives. In column (5) a dummy that equals one if the respondent intends to use contraceptives in the future. In column (6) a dummy that equals one if the respondent does not use contraceptive because of fear to side effects. In column (7) a dummy that equals one if the respondent thinks that aids may be caused by witchcraft. "Traditional beliefs" is a dummy variable that equals one if the respondent declares African traditional religion as his/her main religion. Controls also include age, education, whether the respondent works, marital status, polygamy, and wealth quintiles. Standard errors clustered at the DHS cluster-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table I14: Ancestral beliefs and contraceptive use in the DRC DHS

	Contraceptive use						
	Currently	Traditional	Modern	Children condom	Intention to use	Fear side-effects	Knowledge
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Ancestral Beliefs	-0.0242*	-0.0249**	0.000698	-0.0704***	-0.00576	0.0219**	0.00571
	(0.0130)	(0.0109)	(0.00588)	(0.0272)	(0.0207)	(0.00887)	(0.0234)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	0.185	0.105	0.0800	0.523	0.333	0.0574	0.174
R-squared	0.0965	0.0470	0.0514	0.0617	0.0833	0.0410	0.136
N	24674	24674	24674	20640	19961	5031	24674

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The sample is restricted to women. The outcome variable in column (1) is a dummy that equals one if the respondent was using contraceptives at the time of the survey. In column (2) a dummy that equals one if the respondent was using traditional methods. In column (3) a dummy that equals one if the respondent was using modern methods. In column (4) a dummy that equals one if the respondent thinks that children should be taught about condoms to avoid aids. In column (5) a dummy that equals one if the respondent intends to use contraceptives in the future. In column (6) a dummy that equals one if the respondent does not use contraceptive because of fear to side effects. In column (7) a dummy that equals one if the respondent does not know about any contraceptive method. "Ancestral beliefs" is a dummy variable that equals one if the DHS cluster c of individual i belongs to the ethnic homeland of an ethnic group that traditionally practice ancestor worship as reported in Vansina (1966). Controls also include age, age squared, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

J. Kinship structure, ancestral beliefs and fertility behavior

Table J1: Ethnic-level characteristics by kinship system

	(1)		(2)		(1)-(2)	
	Matrilineal		Patrilineal		Pairwise t-test	
Variable	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean difference
Intensive agriculture	13597	0.000	13421	0.034	27018	-0.034
	24	(0.000)	30	(0.035)	47	
Precolonial development	13597	6.655	13421	6.793	27018	-0.137
	24	(0.206)	30	(0.261)	47	
Political complexity	13597	0.201	13421	0.293	27018	-0.092
	24	(0.112)	30	(0.140)	47	
Malaria Index	14855	12.640	19440	11.624	34295	1.015
	28	(1.067)	36	(1.228)	56	
Ruggedness	14855	51625.918	19440	64575.432	34295	-1.29e+04
	28	(10073.638)	36	(14165.482)	56	
Agricultural suitability	14855	0.314	19440	0.366	34295	-0.052
	28	(0.129)	36	(0.079)	56	
Land quality	14855	5.922	19440	5.678	34295	0.244
	28	(0.186)	36	(0.150)	56	
Rainfall	14855	1492.304	19440	1487.956	34295	4.349
	28	(83.805)	36	(39.736)	56	
Patrilocal	14855	0.032	19440	0.949	34295	-0.917***
	28	(0.021)	36	(0.030)	56	
Altitude	14855	0.475	19440	0.600	34295	-0.125
	28	(0.048)	36	(0.090)	56	
Coast	14855	0.180	19440	0.000	34295	0.180
	28	(0.125)	36	(0.000)	56	
River	14855	0.973	19440	0.852	34295	0.121
	28	(0.019)	36	(0.079)	56	
(Log) Population density	5311	1.428	6830	1.166	12141	0.263
	10	(0.672)	18	(0.333)	26	
High Gods	2462	1.941	10581	1.985	13043	-0.044
	9	(0.067)	18	(0.048)	23	

NOTE. Sample: DRC DHS (2007 and 2003-2014) and Ethnographic Atlas. The value displayed for t-tests are the differences in the means across the groups. "Intensive agriculture" equals one if the society practiced "intensive agriculture" or "intensive irrigated agriculture". "Precolonial development" describes the pattern of settlement of societies, and takes on the values of 1 to 8, with 1 indicating fully nomadic groups and 8 groups with complex settlement. "Political complexity" measures the number of jurisdictional hierarchy beyond local communities, ranging from no levels beyond local community to four levels beyond local community. "Malaria Index" measures the suitability of malaria in the ancestral ethnic homeland. "Ruggedness" is constructed as the average across points on a grid 1 kilometer apart within a country of an index of terrain ruggedness. "Agricultural suitability" is the fraction of land that is suitable for the cultivation of barley, wheat, rye, sorghum, foxtail millet, or pearl millet. "Land quality" is measured as a non-additive combination of climate constraints, soil constraints and terrain slope constraints. "Rainfall" is the average annual precipitation (mm). "Patrilocal" equals one if the dominant post-marital residence is patrilocal. "Altitude" is the mean altitude of the ancestral ethnic homeland. "Coast" equals one if the ancestral ethnic homeland. "River" equals one if there is at least one river in the ancestral ethnic homeland. "(Log) Population density" is the population density of the ancestral ethnic group estimated by Murdock. "High Gods" equals one if the society believes in the existing of a moralizing high god. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level. Standard errors are clustered at the ethnicity level.

Table J2 shows the relationship between fertility preferences and patrilineality, which additionally allows me to explore gender heterogeneity. I find that, on average, individuals in patrilineal societies want about 0.4 (10%) more children (column 1). Moreover, there is great heterogeneity by gender: while the influence of patrilineality is strong for men (0.8 additional children), its influence is close to 0 for women. This is what one would expect according to the simple theoretical framework outlined in Section 5. In a context where the motive to continue the family line is of great importance, patrilineality should only be associated with larger ideal family sizes for men, since one's own children continue the man's lineage only in patrilineal societies. In contrast, children always continue the mother's line, independently of the kinship system. Finally, columns 3 and 4 (as compared to columns 5 and 6) show that the relationship is completely driven by societies with strong beliefs in ancestors.[77]

Table J2: Ancestral beliefs, ideal number of children, and kinship structure in the DRC

	Ideal Nb of Children					
	Full sample		Ancestor Worship = 1		Ancestor Worship = 0	
	(1)	(2)	(3)	(4)	(5)	(6)
Patrilineal	0.351**	0.801***	0.456**	0.874***	-0.0373	0.282
	(0.159)	(0.187)	(0.215)	(0.241)	(0.378)	(0.499)
Female		-0.678***		-0.692***		-1.104***
		(0.0874)		(0.0904)		(0.393)
Patrilineal x Female		-0.666***		-0.621***		-0.459
		(0.121)		(0.135)		(0.435)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	5.544	5.544	5.515	5.515	5.901	5.901
R-squared	0.119	0.120	0.124	0.125	0.102	0.102
N	36141	36141	29318	29318	4367	4367

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is the ideal number of children. The sample is restricted to ethnic groups with ancestor worship in columns (3) and (4) and to ethnic groups without ancestor worship in columns (5) and (6). "Patrilineal" is a dummy variable that equals one if the ethnic group e has patrilineal descent. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, whether the respondent works in agriculture and a dummy variable that equals one if the respondent is in the top 40% of the wealth distribution. Standard errors clustered at the DHS cluster-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

[77]Note that the difference between columns (3)-(4) and columns (5)-(6) is not only statistical significance. Also point estimates are much closer to zero when the ethnic group does not practice ancestor worship.

J.1. *Female fertility and siblings' gender across SSA*

Table J3: Free-riding behavior and fertility across Sub-Saharan Africa

	Number of children ever born		
	(1)	(2)	(3)
Matrilineal x Number of sisters	-0.0179		
	(0.0081)**		
Matrilineal x Number of sisters' children		-0.0274	
		(0.0093)***	
Matrilineal x Number of brothers			-0.00897
			(0.0077)
Controls	Yes	Yes	Yes
Country and Survey Year FE	Yes	Yes	Yes
Mean Y	3.034	4.223	3.034
R-squared	0.641	0.545	0.641
N	572342	56948	572342

NOTE. Data: 120 Country-waves from the DHS. The outcome variable is the total number of children ever born. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, and whether the respondent works in agriculture. Standard errors clustered at DHS cluster-level in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table J4: Ancestral beliefs and free-riding behavior across SSA

	Number of children ever born			
	(1)	(2)	(3)	(4)
Ancestral Beliefs (region)	0.331***	-0.264	0.305***	0.342***
	(0.0441)	(0.212)	(0.0599)	(0.0602)
Ancestral Beliefs x Patrilineal		0.562***		
		(0.215)		
Number of sisters			0.0303***	
			(0.00291)	
Nb of Sisters x Ancestral Beliefs			0.000248	
			(0.0114)	
Nb of Sisters x Ancestral Beliefs x Matrilineal			-0.332***	
			(0.102)	
Number of brothers				0.0319***
				(0.00288)
Nb of Brothers x Ancestral beliefs				-0.0116
				(0.0111)
Nb of Brothers x Ancestral Beliefs x Matrilineal				-0.0838
				(0.0950)
Country and year FE	Yes	Yes	Yes	Yes
Controls	No	No	Yes	Yes
Mean Y	2.919	2.975	3.037	3.037
R-squared	0.602	0.599	0.630	0.630
N	775411	615759	379170	379170

NOTE. Data: 120 Country-waves from the DHS. The outcome variable is the total number of children ever born. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, and whether the respondent works in agriculture. Standard errors clustered at DHS cluster-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.

K. Additional implications

K.1. Ancestral beliefs and social organization

Table K1: Ancestral beliefs, fertility and age-based social organization

	Number of children					
	(1)	(2)	(3)	(4)	(5)	(6)
Ancestral beliefs	0.741**	0.328	0.861**	0.720**	0.895***	0.538*
	(0.354)	(0.248)	(0.389)	(0.330)	(0.306)	(0.308)
Ancestral beliefs x Age-based	-0.113	-0.243	-0.341	-0.674*	-0.792**	-0.505
	(0.404)	(0.283)	(0.455)	(0.379)	(0.370)	(0.355)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	Yes	No	Yes
Mean Y	1.654	1.667	2.379	2.378	2.963	2.964
R-squared	0.0146	0.565	0.0120	0.462	0.0105	0.326
N	1198	1177	775	767	669	662

NOTE. Data: PEW research forum 2008-2009 Survey (Kenya). The sample is restricted to women over the age of 25 in columns 3 and 4, and to women with at least one children in columns 5 and 6. The outcome variable is the total number of children. *Age-based* is an indicator equal to one if the respondent belongs to an aged-based society. The explanatory variable is an indicator equal to one if the respondent answers "yes" to the question "Do you ever participate in traditional African ceremonies or perform special acts to honor or celebrate your ancestors?". Controls include age, age^2, sex, urban/rural place of residence, religion, education, a dummy that equals one if the respondent is married, a dummy variable that equals one if the respondent finds himself/herself in a good economic situation, and a dummy equal to one if the respondent did not have money at some point during the last year to buy food for his/her family. Robust standard errors in parenthesis. *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

L. Background on kinship systems

Two persons are kin when one is descended from the other. Kinship results from the recognition of a *social* relationship between parents and children.[78] A system of kinship and marriage can be looked at as an arrangement which enables persons to live together and co-operate with one another in an orderly social life. Kinship systems determine the set of people to whom an individual is considered related and their social obligations to this group (Radcliffe-Brown and Forde, 1950).

A kinship system that is reckoned by each individual with reference to ascendance and descendance, without distinction between male and female lines, is called a *cognatic system*. Persons are cognatic kin or cognates when they are descended from a common ancestor or ancestress counting descent through males and females (Lorimer, 1954). In a cognatic system, the emphasis in kinship relations rests on the nuclear family. In this type of kinship, kin, as defined by dencendance through both the paternal and maternal sides, is not a cohesive kinship group. It includes individuals on maternal and paternal sides, at each ascending and descending juncture, who may not be related to one another [79]

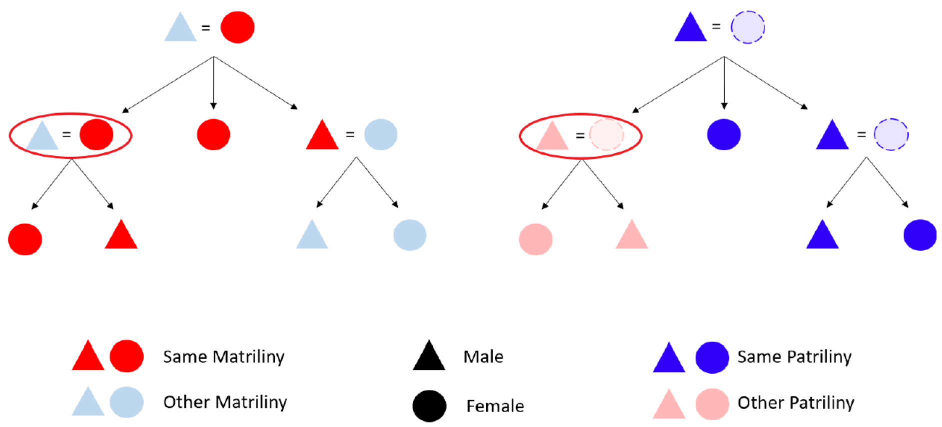
On the other hand, descent groups in which one belongs exclusively to one's father's or mother's line are called *unilineal systems*. In unilineal descent groups, one's descent is traced either exclusively through male ancestors (patrilineal descent groups), or exclusively through female ancestors (matrilineal descent groups). These groups form a cohesive and continuing kinship structure. In fact, the unity of the sibling group is preserved (especially as regards those who by virtue of their sex are part of a continuing lineage). A group of brothers, or a group of sisters, acquire hereditary rights and transmit them to their descendants of the same sex. There is, however, an important difference between patrilineal and matrilineal descent systems that matters for the analysis in this paper, known as asymmetric marital allegiances (Berggreen and Gokmen, 2023). In pa-

[78]The stress of the word social is key to distinguish kinship from consanguinity (physical relationship). In fact, an illegitimate child has a physical father but may not have a social father (he does not belong to the same kin group as his biological father). Similarly, an adopted child may be part of the same kin group as his adoptive parents while they are not consanguineous.

[79]As Lorimer (1954) notes: "I may feel myself to be related to the children and grandchildren of my paternal grandfathers brothers and sisters, and to those of my paternal grandmother, my maternal grandfather, and my maternal grandmother; but these four sets of relatives have no necessary relationship to one another".

trilineal societies, the children belong to the father's lineage, and a patrilineal daughter who marries becomes part of her husband's lineage. Then, both spouses and children are part of the same kin group in patrilineal societies. In matrilineal societies, children belong exclusively to the mother's kin group, and a patrilineal son who marries maintains his birth lineage (in fact, inheritance often passes from the maternal uncle to his sisters' children). Figure L1 summarizes these differences:

Figure L1: Matrilineal and patrilineal kinship systems, from Berggreen and Gokmen (2023)



Source: Berggreen and Gokmen (2023). The figure shows the kinship structure of both patrilineal and matrilineal societies. The sign = symbolizes marriage, while the bracket over the triangle/circle indicates siblingship.

Table L1: Ancestral beliefs, free-riding behavior, and marital outcomes

	Age at first marriage (log)				Age at first birth (log)			
		Ancestors=1	Ancestors=0	Ancestors=1		Ancestors=1	Ancestors=0	Ancestors=1
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Number of sisters	-0.00472***	-0.00506***	-0.00176		-0.00404***	-0.00380***	-0.00510**	
	(0.00110)	(0.00127)	(0.00273)		(0.000993)	(0.00113)	(0.00245)	
Matrilineal x Sisters	0.00288*	0.00316*	-0.00216		0.00243*	0.00259	-0.00589	
	(0.00155)	(0.00171)	(0.00461)		(0.00144)	(0.00158)	(0.00483)	
Number of brothers				-0.00155				-0.00227**
				(0.00121)				(0.00112)
Matrilineal x Brothers				-0.000849				-0.00152
				(0.00174)				(0.00156)
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Y	2.872	2.875	2.856	2.875	2.937	2.938	2.928	2.938
R-squared	0.110	0.120	0.0595	0.119	0.108	0.114	0.0862	0.114
N	19967	15936	2655	15936	19540	15655	2563	15655

NOTE. Data: Demographic and Health Surveys of DRC (2007 and 2013-2014). The table reports OLS estimates. The outcome variable is log of the age at first marriage in columns 1-4 and the log of the age at first birth in columns 5-8. The sample is restricted to women. The sample is restricted to ethnic groups with ancestor worship in columns 2,4,6 and 8, and to ethnic groups without ancestor worship in columns 3 and 7. "Matrilineal" is a dummy variable that equals one if the ethnic group e has matrilineal descent. "Number of sisters" is the number of sisters that the respondent has ever had. "Number of brothers" is the number of brothers that the respondent has ever had. The total number of alive siblings is always included as control variable. Controls include age, age squared, gender, a dummy variable that equals one if the respondent is catholic, single years of education, whether the DHS cluster is urban, whether the respondent works, and whether the respondent works in agriculture. Standard errors clustered at the DHS cluster-level in parenthesis. *** for < 0.01, ** for < 0.05, * for < 0.1.